Selecting Cloud and Data Infrastructure for a Smart Factory

Technical report: data volume, cloud comparison, decision matrix, PostgreSQL design, Azure reference architecture, edge-versus-cloud analysis and analytics for a 100-machine CNC fleet

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Repository: https://github.com/TirthankerSingh/industrial-data-platform

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Recommended platform: Microsoft Azure (Central India), edge-first design

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Executive summary

A manufacturer runs **100 CNC machines in three plants**. In this report the scenario is an automotive Tier-1 supplier at Chakan (Pune), Oragadam (Chennai) and Manesar (Gurugram). The report designs and justifies the cloud-data architecture that will carry its sensor, production and machine-control data.

What makes this report evidence-based.

- Every quantitative claim rests on **four public datasets** that were actually downloaded and processed:
  - two **real** industrial datasets: Bosch CNC vibration, with 1,702 machining operations at 2 kHz [1], and the UMich CNC mill, with 18 experiments [2];
  - two **benchmarks**: Microsoft Azure Predictive Maintenance, simulated [3], and AI4I 2020, synthetic [4].
- On top of these were built:
  - a PostgreSQL database with 876,100 machine-hours;
  - an MQTT edge gateway with store-and-forward;
  - four machine-learning models;
  - an 8-page dashboard.

31.5 bn	**143×**	**4.22 / 5**	**0 lost**
values/year, 1 Hz fleet	edge feature reduction (real data)	Azure weighted score	of 482 MQTT messages in outages

Key results.

1. **Data volume.**
   - The 1 Hz fleet produces 1,000 values/s, which is **31.536 billion a year**.
   - One 100 Hz vibration sensor adds 3.15 billion a year; fitting one to every machine adds 315 billion, ten times the whole 1 Hz fleet.
   - At the real Bosch rate (2 kHz × 3 axes), raw vibration would need 96 Mbps and 378 TB a year.
   - Lossless compression achieves only 5× on vibration. Edge feature extraction achieves **143×** while keeping the fault signal (ROC-AUC 0.98).
2. **Cloud comparison.** Official sources were checked on 2026-10-21.
   - Several IIoT services are retired or in maintenance: GCP IoT Core, Azure Time Series Insights, AWS IoT Analytics and AWS IoT Events [5][6][7][8].
   - Azure offers the most complete current shop-floor-to-dashboard chain. AWS has the best-documented edge buffering. GCP has no first-party IIoT layer.
3. **Decision matrix.** Across 10 weighted criteria the scores are **Azure 4.22**, AWS 3.92 and GCP 3.57.
   - Azure wins all ±5-point weight shifts and 88.7% of 10,000 Monte-Carlo runs.
   - AWS overtakes only when cost carries at least 35% of the weight.
4. **Cost.**
   - An edge-batched design costs ₹33k–80k per month.
   - Sending one MQTT message per reading would cost ₹14–30 lakh per month, so **edge batching is the largest cost lever**.
5. **Recommendation: Microsoft Azure, Central India, with an edge-first design.**
   - Each plant runs IoT Operations, reading OPC UA from the Fanuc and Siemens controllers, plus a durable 7-day outbox.

- Data flows through IoT Hub / Event Hubs into Fabric Eventhouse and ADLS, then PostgreSQL Flexible Server, Azure ML and Power BI.
- The scenario reasons are brownfield OPC UA, the Microsoft tooling already used in the plants, a Pune region next to Chakan, and an edge design that removes Azure's offline-time and cost weaknesses.

1. Introduction and industrial scenario

The task. The manufacturer needs infrastructure for monitoring, analytics, storage, dashboards, AI/ML and remote access across 100 CNC machines. The assignment asks for seven things:

- the data volume, and why high-frequency data needs edge processing (Part A);
- a 13-criterion cloud comparison (Part B);
- a weighted decision matrix and recommendation (Part C);
- a SQL database with eight analytical queries (Part D);
- a GitHub repository (Part E);
- a mapped cloud architecture (Part F);
- an edge-versus-cloud analysis (Part G).

The scenario. To make the recommendation concrete, this report uses an illustrative automotive Tier-1 supplier in India (Table 1). The fleet uses Fanuc 0i/31i and Siemens SINUMERIK 828D/840D sl controllers. Each machine carries 10 sensor streams at 1 Hz:

- spindle temperature, coolant temperature, spindle vibration (RMS) and spindle RPM;
- spindle motor current, feed-axis load, hydraulic pressure and coolant flow;
- power consumption and door/cycle status.

It also carries one spindle accelerometer: 100 Hz in the assignment calculation, and 2 kHz × 3 axes in the real Bosch data. The plants run 24×7 on three shifts (A 06–14, B 14–22, C 22–06).

Table 1: Plants, machines and part families (scenario assumptions; data/reference/)

Plant	Shop floor	Machine IDs	Machine types	Part families (nominal cycle time)	Real data anchoring it
P1 Chakan, Pune	Powertrain machining	1–40	24 HMC, 16 VMC	Engine block (14 min), cylinder head (7.5 min), crankcase (11 min)	Bosch vibration [1]
P2 Oragadam, Chennai	Transmission & driveline	41–75	15 turning, 10 hobbing, 10 grinding	Gear shaft (3.5), helical gear (4.5), differential housing (6), finish-ground shaft (2.5)	AI4I temperature/torque/wear [4]
P3 Manesar, Gurugram	Chassis & brakes	76–100	17 turning, 8 VMC	Brake disc (2), steering knuckle (5), wheel hub (3)	UMich spindle current [2]

Nominal spindle speed ranges (scenario assumptions):

Machine type	RPM range
HMC	2,000–8,000
VMC	3,000–10,000
Turning	800–3,500
Hobbing	300–1,500
Grinding	1,500–4,500

Rated spindle current follows from rated power: $I = P / (\sqrt{3} \cdot 415\ V \cdot 0.85)$.

2. Datasets and provenance

Table 2: Datasets used, their nature and licence

Dataset	Nature	Size used	Licence	Role
Bosch CNC Machining [1]	**REAL** production plant	1,702 files, 3 machines, 2 kHz × 3 axes, 24.8 h, 70 bad	CC-BY-4.0	Edge features, compression, anomaly models, vibration scale
UMich SMART CNC Mill [2]	**REAL** CNC machine	18 experiments, 25,286 rows × 48 signals at 100 ms	as published	Current calibration, reject model, tool-wear models, MQTT replay
Azure PdM [3]	**BENCHMARK**, simulated by Microsoft	100 machines × 8,761 h, errors, maintenance, failures	as published	Fleet backbone, 24-h failure prediction
AI4I 2020 [4]	**BENCHMARK**, synthetic	10,000 rows, 5 failure modes	CC-BY-4.0	Temperature model, failure-mode classifier

How the fleet table is built.

- The PostgreSQL fact table uses the Azure telemetry as its **backbone**. Machine IDs 1–100 equal the Azure machineIDs, so every row is traceable.
- All other values are **DERIVED** with documented rules; everything is in `data/DATA_PROVENANCE.md`.
- The raw source tables are loaded unchanged into the `raw` schema, so each derived value can be traced back.

Derivation rules.

- **Timestamps:** shifted by 3,654 days (522 weeks, so weekdays are preserved) to 2 Jan 2025 – 2 Jan 2026 IST.
- **RPM:** a linear rescale of `rotate` into each machine type's range.
- **Spindle current:** `volt` is mapped onto the real UMich cutting-load band (22/31/41% of peak current).
- **Vibration (mm/s RMS):** an affine map. The fleet median goes to 1.8 mm/s (centre of ISO 10816-3 zone B [9]); the pre-failure level goes to the Bosch-measured fault ratio (90th percentile 1.68×).
- **Temperature:** an AI4I regression with plant ambient as input (cross-validated R^2 0.77).
- **Alarms:** each Azure event type is named after the signal that shifts in the 24 h before it.
- **Status:** derived from alarms, maintenance, holidays and low-load periods.
- **Production:** running minutes ÷ cycle time.
- **Rejects:** a logistic model using UMich's worn-tool odds ratio of 13 and AI4I coefficients, solved for a 1.5% fleet rate.

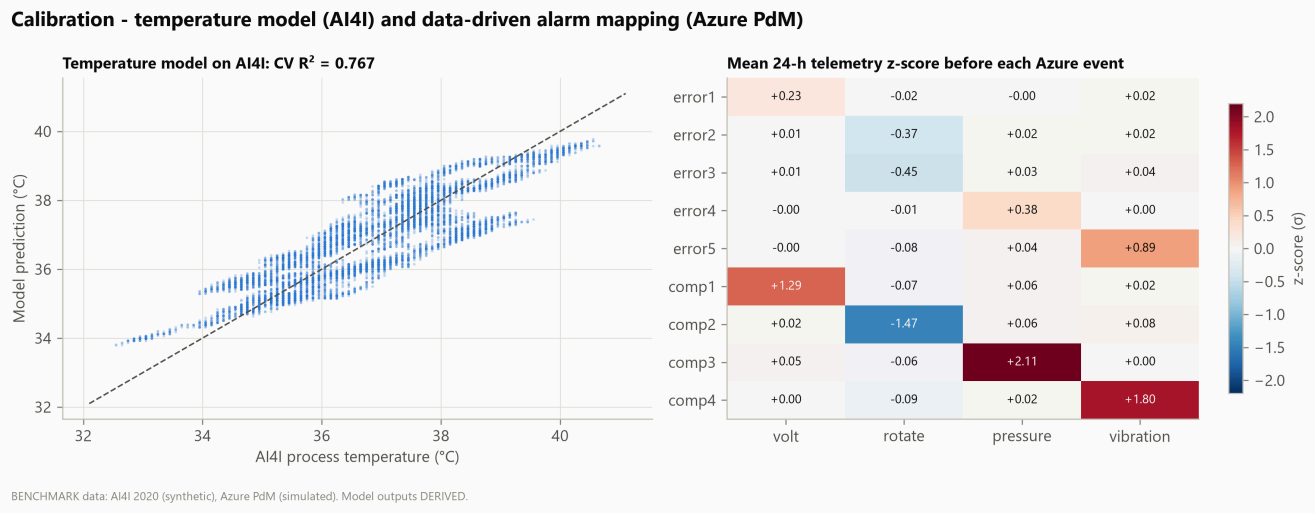


Figure 1: Calibration: AI4I temperature model (left) and data-driven alarm naming (right), where each Azure event is named after the telemetry signal that moves in the 24 h before it

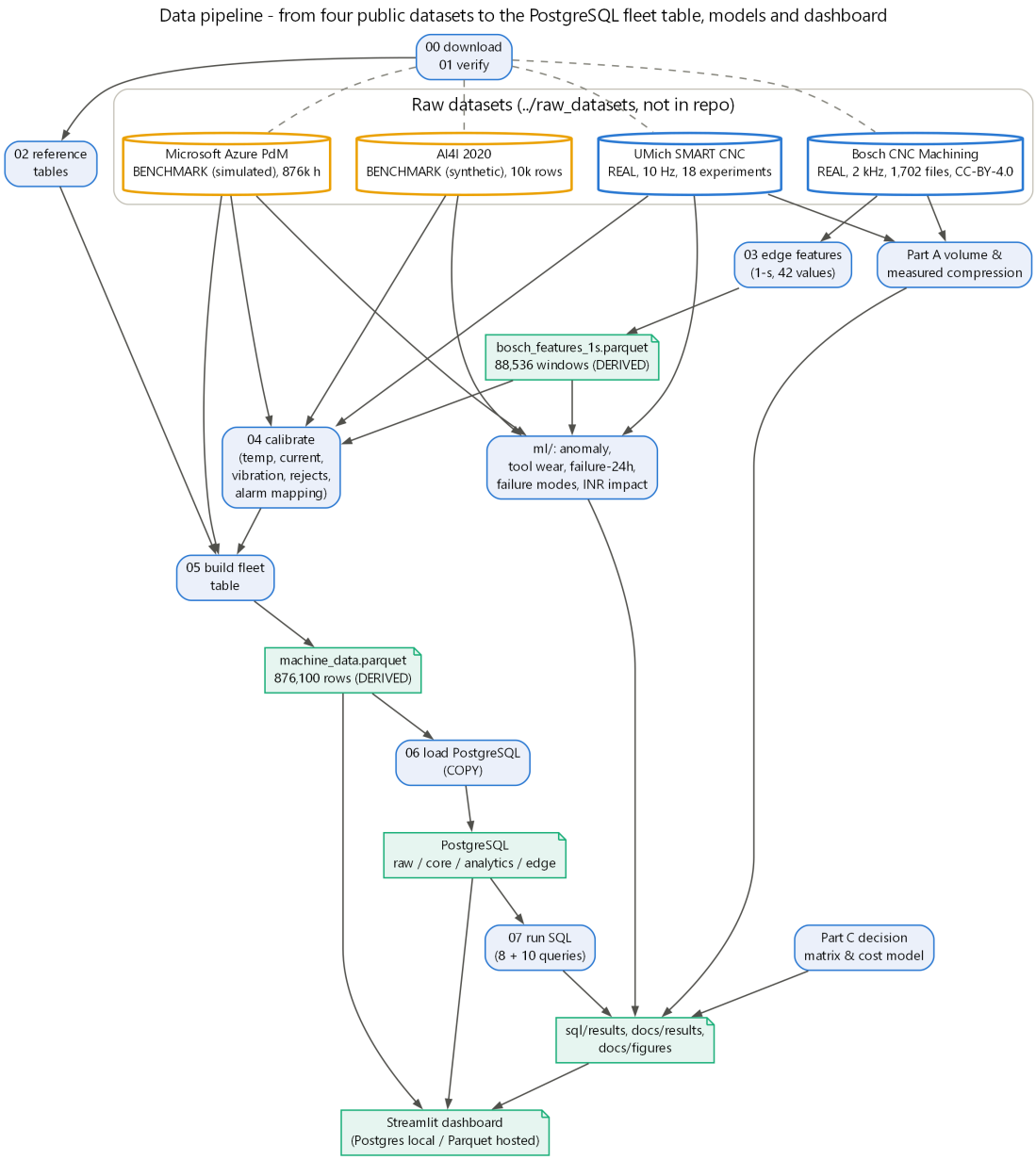


Figure 2: End-to-end data pipeline: four datasets, numbered scripts, outputs

3. Part A: Industrial data volume

A.1 Measurements: 100 machines × 10 sensors × 1 Hz

Period	Seconds	Measurements
per second	1	**1,000**
per hour	3,600	**3,600,000** (3.6 million)
per day	86,400	**86,400,000** (86.4 million)
per month (30 d)	2,592,000	**2,592,000,000** (2.592 billion)
per year (365 d)	31,536,000	**31,536,000,000** (31.536 billion)

A.2 One vibration sensor at 100 Hz

Period	One sensor, one machine	Same sensor on all 100 machines
per second	**100**	10,000
per hour	**360,000**	36,000,000
per day	**8,640,000** (8.64 million)	864,000,000
per month (30 d)	**259,200,000** (259.2 million)	25,920,000,000
per year (365 d)	**3,153,600,000** (3.1536 billion)	315,360,000,000

The "shock number".

- One 100 Hz vibration sensor on one machine produces **3.15 billion readings a year**. That is ten times all 10 slow sensors on that machine, and as much as the 1 Hz streams of **ten whole machines**.
- Fitting one to each of the 100 machines gives **315 billion readings a year, ten times the whole 1 Hz fleet**.
- The real Bosch accelerometers sample at **2 kHz on 3 axes**. Streaming that raw from 100 machines would mean **18.9 trillion values a year** (figure `part_a_volume_per_year.png`).

A.3 Storage and bandwidth

Byte-size assumptions:

- **Binary record:** 8 B timestamp (int64) + 8 B value (float64) + 2 B machine ID + 2 B sensor ID = **20 B**.
- **Value-only:** 8 B, with timestamps implicit at a fixed rate.
- **JSON over MQTT**, sizes **measured** by serialising realistic messages:
  - one value per message = 125 B payload, 219 B on the wire including topic, MQTT header and TCP/IP;
  - 10 sensors batched per second = 38 B per value;
  - a 100 Hz one-second block = 8.25 B per value.

Scenario (20 B binary record)	Bandwidth	Per day	Per year
Fleet 1 Hz (100 × 10 × 1 Hz)	0.16 Mbps	1.73 GB	0.63 TB
One 100 Hz sensor	0.016 Mbps	0.17 GB	0.06 TB
Fleet 100 Hz (100 × 1 × 100 Hz)	1.60 Mbps	17.3 GB	6.31 TB
Design point (10 × 1 Hz + 1 × 100 Hz per machine)	1.76 Mbps	19.0 GB	6.94 TB
Raw 2 kHz × 3 axes on 100 machines	**96 Mbps**	**1,037 GB**	**378 TB**

Encoding matters as much as the sample rate. With one JSON message per reading, the 1 Hz fleet alone needs 1.75 Mbps, 11× the binary figure, and the raw 2 kHz case exceeds 1 Gbps (figure `part_a_bandwidth_by_encoding.png`).

A.4 Measured compression (real data, not assumed)

Each dataset was written to disk in six formats with pyarrow (figure `part_a_measured_compression.png`):

Format	UMich CNC 10 Hz, 48 signals (REAL)	Bosch 2 kHz vibration (REAL)	Azure PdM hourly (BENCHMARK)
CSV (baseline)	6.5 MB = 1.0×	285 MB = 1.0×	77.5 MB = 1.0×
CSV + gzip	4.7×	4.1×	2.4×
Parquet, uncompressed	4.8×	2.8×	2.5×
Parquet + Snappy	5.8×	3.3×	2.5×
Parquet + Zstd (level 3)	**6.9×**	**5.0×**	**2.7×**
Parquet + Zstd (level 9)	7.1×	5.0×	2.7×

Reading:

- Slow, structured CNC signals compress about 7×, because columns repeat values and change smoothly.
- Raw vibration compresses only about 5× against CSV, and barely beyond its own binary size. **Vibration is mostly broadband noise, and lossless compression cannot remove noise.**
- That is why feature extraction, not compression, is the right tool for high-frequency data.

A.5 Edge feature extraction on real data (key result)

What was done.

- All **1,702** Bosch files were processed: 24.8 h of real 2 kHz tri-axial data, 178.6 M samples per axis.
- Each was cut into **88,536 one-second windows**.
- Per axis and window: mean, RMS, peak, peak-to-peak, crest factor, kurtosis, skewness, five FFT band-energy fractions, dominant frequency and spectral centroid. That is **42 values per second instead of 6,000 samples**.

Result	Value
Reduction per machine-second	24,000 B → 168 B = **143×**
Measured on the full dataset	**120×** vs raw float32; **53×** vs the compressed HDF5 files on disk
Fleet uplink (100 machines)	**19.2 Mbps → 0.13 Mbps**
Fleet storage per year	**75.7 TB → 0.53 TB**
Extraction speed	40 s for 24.8 h of data on a laptop CPU, about 2,200× faster than real time
Is the fault signal kept?	Yes. Single features separate good from bad operations: ROC-AUC up to **0.82** for the Y-axis 150–300 Hz band. An Isolation Forest trained only on good operations reaches **0.98**, and **0.98** when trained on 2019 data and tested on 2021.

Figures:

- `edge_waveform_vs_features.png`: raw waveform vs 1-s RMS and kurtosis trend.
- `edge_spectrogram_good_vs_bad.png`: the bad operation shows a strong ~250 Hz line and lacks the normal spindle ramp-up.
- `edge_feature_space_pca_tsne.png`: good and bad separate in feature space, and the per-machine view shows the domain shift.
- `edge_top_features_boxplot.png` and `edge_data_reduction.png`.

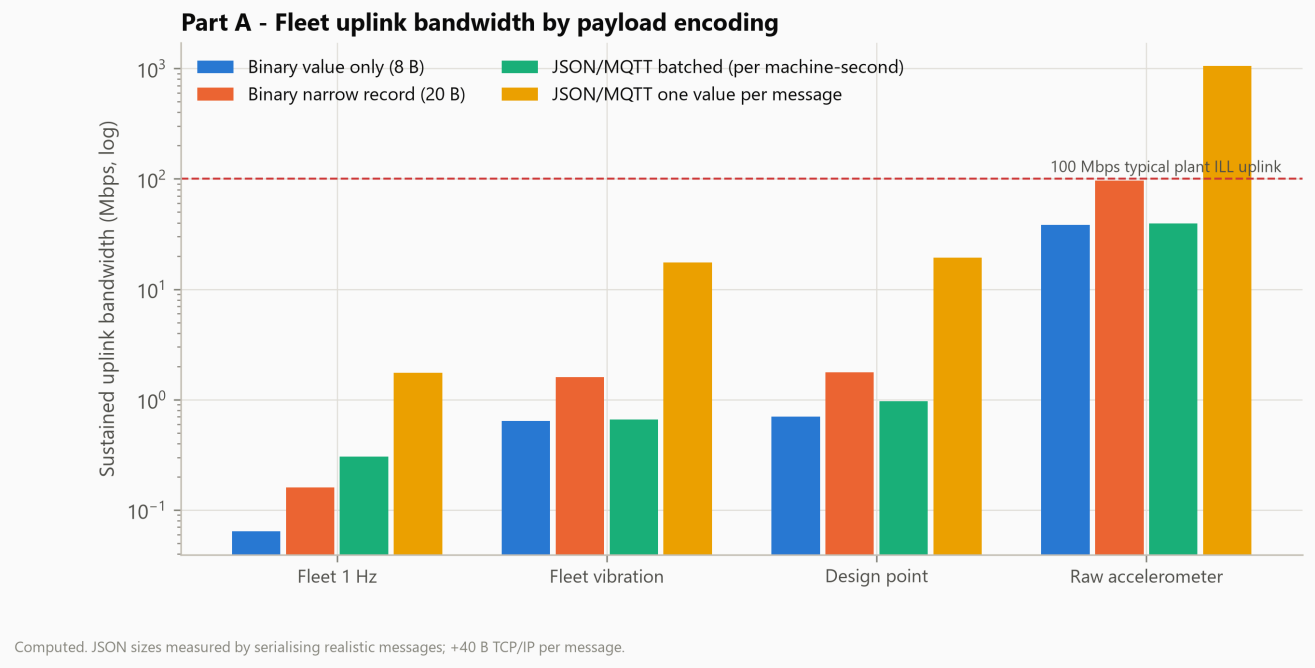


Figure 3: Fleet uplink bandwidth by payload encoding; the dashed line is a typical 100 Mbps plant leased line

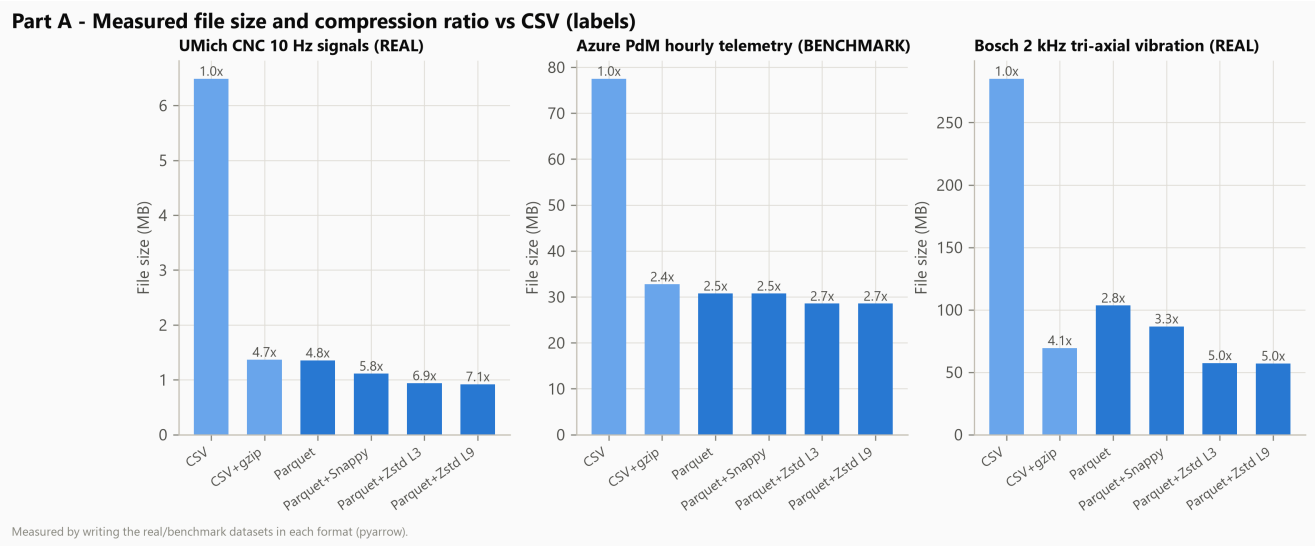


Figure 4: Measured file size and compression ratio vs CSV for real UMich, real Bosch and benchmark Azure data

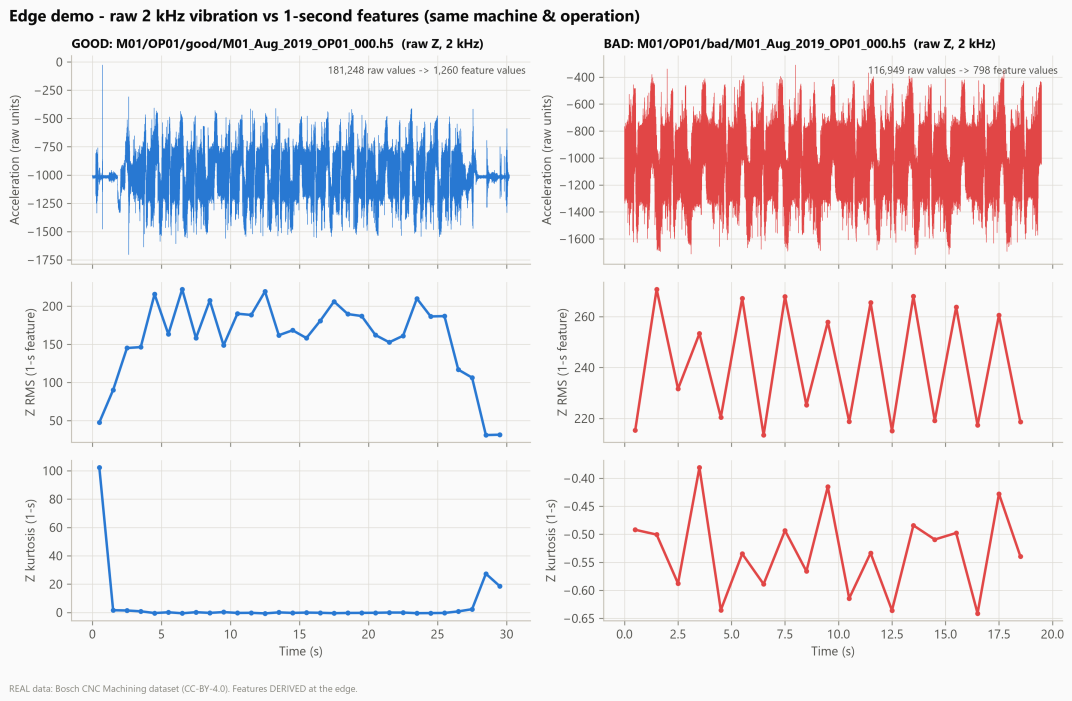


Figure 5: Raw 2 kHz Z-axis vibration vs the 1-second RMS and kurtosis features computed at the edge (same machine and operation)

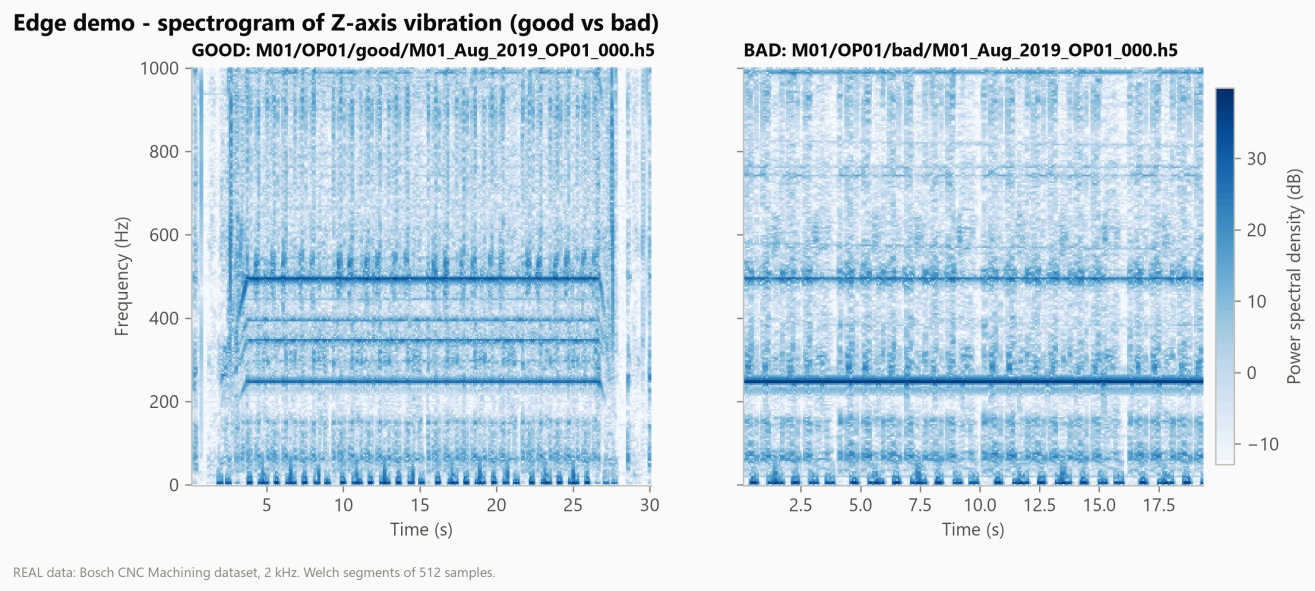


Figure 6: Spectrogram of a good and a bad operation: the bad operation shows a strong ~250 Hz line and lacks the normal spindle ramp-up



Figure 7: Good vs bad operations in feature space: PCA of file-level features, t-SNE of 1-s windows, and the per-machine domain shift

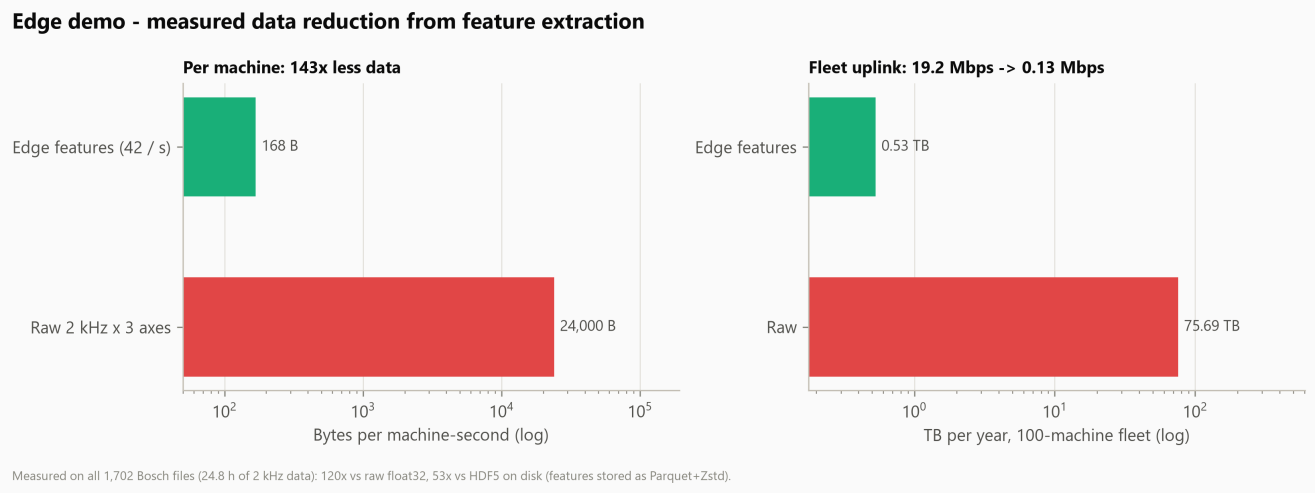


Figure 8: Measured data reduction from edge feature extraction

A.6 Discussion

Why high-frequency vibration creates storage and transmission challenges.

- Vibration has to be sampled at 2–10× the highest frequency of interest (Nyquist). Bearing and tool faults show up between 100 Hz and several kHz, which forces kHz rates.
- **Volume:** a single 100 Hz channel on every machine generates ten times the readings of the entire 1 Hz fleet. At the real 2 kHz × 3 axes it is 600,000 values/s for 100 machines.
- **Bandwidth:** 96 Mbps sustained would take a plant's whole internet link, which is typically 50–100 Mbps of leased line shared with ERP, video and office traffic.
- **Storage:** 378 TB/year raw, and noise does not compress (A.4).
- **Cost:** per-message cloud pricing explodes when each reading is its own message. ₹14–30 lakh/month against ₹0.4–0.8 lakh batched (`cost_model.md`).
- **Ingestion and processing:** time-series databases and dashboards slow down at billions of rows. Most of those rows are noise that nobody queries.

Why raw high-frequency data should not always go to the cloud.

- The decision-relevant information is far smaller than the signal itself. RMS, kurtosis and band energies carry the fault signature (AUC 0.98 here), so shipping raw data pays 143× more for very little extra information.
- Protective actions (spindle stop, high-vibration alarm) cannot wait for a WAN round-trip, or for an internet outage to end.

- Raw OT data leaving the plant enlarges the attack surface and the data-residency scope.
- Raw data is still useful **occasionally**: for root-cause analysis and for training new models. Keep a short ring buffer at the edge and upload raw windows on demand, for example the 10 s around an anomaly.

The role of edge processing.

- The edge gateway sits beside the machines (Purdue L3). It converts protocols, filters noise, checks data quality, extracts features, batches messages, runs alarms and local inference, and buffers data while the WAN is down.
- In this project's MQTT demo, the gateway's outbox delivered all 482 messages through a subscriber outage and a WAN outage, with 0 lost and 0 duplicated.

Compression, aggregation and feature extraction: three different tools.

Technique	What it keeps	Typical gain (measured here)	Use for
Lossless compression (Parquet+Zstd, gzip)	Every bit	5–7× (2.7× for already-aggregated data)	Archives, the data lake
Aggregation / downsampling (1-min mean/min/max, 1 Hz)	Trends, levels	60× for 1 Hz → 1 min	Dashboards, OEE, slow sensors
Batching (one message per machine-second)	Every value; fewer messages	35–70× lower ingestion cost	All telemetry
Event-based / report-by-exception (send on change > deadband)	Changes only	Large for slow-changing signals (status, temperature)	Status, setpoints, alarms
Feature extraction (RMS, kurtosis, FFT bands, crest factor)	The signal's diagnostic content	**143×**	High-frequency vibration, current signatures

Combined design for this factory:

- 1 Hz sensors are batched per machine-second and sent as is.
- Status and alarms are sent by exception.
- Vibration becomes 42 features per second, with raw windows kept for 7 days at the edge.
- Everything is stored as Parquet+Zstd in the lake.
- Result: about 0.69 Mbps for the fleet and 0.4 TB per year of compressed history.

4. Part B: Cloud platform comparison

Scenario. An automotive Tier-1 supplier runs 100 brownfield CNC machines (Fanuc and Siemens SINUMERIK controllers) in three plants: P1 Chakan (Pune), P2 Oragadam (Chennai) and P3 Manesar (Gurugram). Each machine produces 10 sensor streams at 1 Hz plus one 2 kHz vibration channel that is reduced at the edge. Part A shows the edge cuts this by 143×.

What the platform must do.

1. Read OPC UA data from two controller families.
2. Survive the plants' internet outages without losing data.
3. Keep data in India.
4. Feed a PostgreSQL / analytics / ML / dashboard stack.
5. Be run by a small OT and IT team at a cost that is defensible in INR.

Method.

- Facts come from official documentation and pricing pages accessed on 2026-10-21. Each fact is cited as `[S#]` and listed in `sources.md`; detailed notes are in `cloud_research_notes.md`.

- Text marked *Assessment* is engineering judgement, not a sourced fact.

B.1 Service status check (2026-10-21): what is no longer available

Several IIoT services that older comparisons still recommend have been retired or put into maintenance mode. This design does not use any of them.

Service	Status	Consequence for this design
Google Cloud IoT Core	**Retired 2023-08-16** [5]	GCP has no first-party device-connectivity service. A broker or partner layer is required (B.2) [10].
Azure Time Series Insights	**Retired 2024-07-07** [6]	Time-series analytics moves to Azure Data Explorer / Fabric Eventhouse [6].
Azure IoT Central	GA; **retirement announced for 2029-09-20** [11] (*not fully confirmed: read via a mirror*)	No new build on IoT Central. Use IoT Hub / IoT Operations instead.
AWS IoT Analytics	**End of support 2025-12-15** [7]	AWS's own migration path is Kinesis + Glue + Athena [12].
AWS IoT Events	**End of support 2026-05-20** [8]	Alarm logic must be rebuilt elsewhere (edge rules, Lambda, SiteWise alarms).
AWS IoT SiteWise Monitor	**Maintenance mode, closed to new customers 2025-11-07** [13]	Dashboards move to Amazon Managed Grafana [13].
AWS SiteWise Edge Data Processing Pack	**Maintenance mode** [14]	Edge data *collection* continues [14]; edge processing needs partners or open source [15].
AWS IoT Greengrass V1	**End of support 2026-10-07** [16]	Greengrass V2 only.
Amazon Timestream for LiveAnalytics	**Closed to new customers 2025-06-20** [17]	Timestream for InfluxDB instead [18].
Azure IoT Edge 1.5 LTS	**End of support 2026-11-10**; 1.6 LTS supported to 2028-11-14 [19]	Any Azure edge design must target 1.6 LTS or IoT Operations.
Microsoft Fabric RTI, Azure IoT Operations, Event Grid MQTT	**GA** [20][21][22]	These are the current Azure building blocks.

B.2 How each platform connects the shop floor today

- **AWS.**
  - **Edge:** SiteWise Edge runs on Greengrass V2 or on Siemens Industrial Edge. It reads OPC UA and MQTT, and keeps collecting during internet outages, syncing when the link returns [23].
  - **Cloud:** IoT Core (MQTT) and SiteWise (asset models, computed metrics, alarms) [14][24].
  - **Weakness:** the portfolio around it has thinned (B.1). The team has to assemble dashboards (Grafana), stream processing (Kinesis) and query (Athena) itself.
- **Azure.**
  - **Edge:** Azure IoT Operations (AIO) runs on Arc-enabled Kubernetes at the plant. It provides an OPC UA connector with asset discovery into Azure Device Registry, certificate management, high availability and write-back [25], plus an MQTT broker and edge data flows [26].
  - **Cloud:** IoT Hub / Event Grid MQTT [21][27], then Fabric Real-Time Intelligence (Eventstreams → Eventhouse → Real-Time Dashboards, Activator) or Azure Data Explorer [22][28].
  - **Caveats:** AIO can run offline for **at most 72 h**, and its disk buffer is off by default [29][26]. AIO's Arc-connection regions include **no India region**, although cloud resources can still sit in Central India [20].
- **GCP.**
  - **No first-party IIoT layer.** Google documents three patterns: a self-run MQTT broker on GCE/GKE, a third-party IoT platform, or devices straight to Pub/Sub [10][30].

- **Manufacturing offering:** Manufacturing Connect (MC), which is **built, sold and supported by Litmus** (270+ protocols including OPC UA), plus the Manufacturing Data Engine (MDE), deployed into the customer's own project on Pub/Sub, Dataflow, BigQuery and Bigtable [31][32].
- **Consequence:** the edge is a third-party contract.

Controller reality, for all three clouds.

- Fanuc controllers need FANUC's FASOPC OPC server or a FOCAS collector [33].
- SINUMERIK 840D sl / 828D need the *Access MyMachine / OPC UA* licence [34].
- So every option needs either per-machine controller licences or an edge product with native FOCAS/S7 drivers [14][31].

B.3 Thirteen-criterion comparison: how each platform serves this factory

Criterion	AWS	Microsoft Azure	Google Cloud
Industrial IoT capability	SiteWise models each CNC as an asset with computed OEE-style metrics and alarms in the cloud [14]; SiteWise Edge reads OPC UA from the plant network [23]. Gaps: IoT Events retired [8], Monitor in maintenance [13], so P1–P3 alarm rules and dashboards must be rebuilt in other services.	IoT Operations gives a Kubernetes-hosted OPC UA connector with asset discovery, certificates and write-back [25]. This maps directly to 100 machines on three shop-floor networks, registered once in Azure Device Registry. IoT Central is retiring [11], so the design avoids it.	No first-party IIoT service since IoT Core retired [5][10]. The factory would buy Litmus-based Manufacturing Connect [31] or run its own broker. Workable, but every plant's connectivity depends on a partner.
Data ingestion	IoT Core MQTT at $1.05 per million messages, metered in 5 KB steps [24][35]. One 1-s batch per machine (≈0.9–1 KB, Part A) is one message, about 259 M messages/month for the fleet. Kinesis for high-rate streams [36].	IoT Hub units are prepaid daily quotas (S1: 400k msg/day, 4 KB meter) [37]. The fleet's 8.64 M batched messages/day need 2 × S2 units (₹66,300/month), a predictable but significant flat price []. Event Hubs / Event Grid MQTT for broker-style fan-in [21][].	Pub/Sub billed per TiB ($40/TiB) [38]. Cheap for the edge-reduced stream (~0.5 TB/year), but MQTT termination needs a self-run broker VM or a partner [30].
Storage options	S3 Standard $0.025/GB-month in Mumbai [39] as the data lake; SiteWise hot/warm tiers [40]; Timestream for InfluxDB (LiveAnalytics closed to new customers) [17][18]. Bosch-style raw archives fit S3 lifecycle tiers.	ADLS Gen2 Hot $0.020/GB-month in Central India []; Eventhouse/ADX for hot time-series with KQL [22][28]; OneLake in Fabric []. The lake sits in Pune, about 30 km from P1 Chakan.	Cloud Storage and BigQuery in asia-south1 (Mumbai) and asia-south2 (Delhi, near P3 Manesar) [41][42]. MDE lands data in BigQuery / Bigtable [32].
Stream processing	Kinesis Data Streams plus consumers (Lambda / Flink) [36]; SiteWise computed properties for per-machine metrics [14]. Rolling OEE or vibration baselines need custom code.	Fabric Eventstreams: no-code filters, windowed aggregation, dedup and routing [22]; ADX update policies and materialized views [28]; AIO data flows at the edge [26]. This covers 1-min OEE and alarm routing without custom services.	Dataflow (Apache Beam) in both India regions [43]: powerful, but code-first, which is a heavier skill demand for a plant team.
SQL and structured data	RDS / Aurora PostgreSQL [44]. Mumbai list price not retrievable [45]. The schema in `sql/` runs unchanged.	Azure Database for PostgreSQL Flexible Server, priced in Central India (B2s $0.098/h, D2ds_v5 $0.251/h) []. The schema and queries in `sql/` run unchanged.	Cloud SQL for PostgreSQL in asia-south1/2 [46][47]; BigQuery for large scans.
Analytics	Athena / Glue over S3 [12]; Managed Grafana for SiteWise dashboards [13]; QuickSight.	KQL anomaly and forecasting functions in ADX/Eventhouse [28], Real-Time Dashboards, and Power BI native in Fabric [22]. Plant managers already using Excel/Power BI get dashboards without new tools.	BigQuery, serverless SQL at $6.25/TiB scanned [48]: the strongest ad-hoc analytics engine for fleet-wide history.

Criterion	AWS	Microsoft Azure	Google Cloud
AI/ML tools	SageMaker (general knowledge); SiteWise Assistant GenAI add-on [40].	Azure ML; Fabric notebooks; native anomaly detection in Eventhouse [22] [28]. The Bosch Isolation Forest and the Azure 24-h failure model (Part ML) can be registered and served from Azure ML.	Vertex AI and BigQuery ML (general knowledge). MDE was designed to feed Vertex AI. Strong for data-science-heavy teams.
Edge computing	Greengrass V2 + SiteWise Edge with explicit outage buffering [23][16]; can run on Siemens Industrial Edge. Edge processing pack in maintenance [14].	IoT Operations on Arc-enabled K8s (K3s / AKS Edge Essentials) [20]; IoT Edge 1.6 LTS [19]. Needs a small K8s cluster per plant. 72-h offline limit [29].	MC (Litmus Edge) [31]; Google Distributed Cloud connected for on-prem GKE [49] (India availability unknown).
Security	X.509 per device in IoT Core; IoT Device Defender (general knowledge). SiteWise is not in scope of AWS compliance programmes [50]. AWS is an ISASecure member, which is not a product certification [51].	AIO uses production PKI for OPC UA (≥2048-bit, SHA-1 rejected), X.509 broker auth, managed identities and signed images [29]; Defender for IoT gives agentless OT network monitoring, also air-gapped [52]. Caveat: its cloud-connected mode routes non-EU data to East US [52].	Pub/Sub can pin message storage to India regions [53]; edge security is the partner's (Litmus) responsibility [31].
Scalability	IoT Core tiers past 5 B msg/month [35]: 10× fleet growth is a pricing-tier event, not a redesign.	IoT Hub S3 unit = 300 M msg/day [37]; ADX ingests millions of events/s [28]. 1,000 machines = add units.	Pub/Sub autoscales; brokers rated for "millions" of devices [10].
Integration	OPC UA via SiteWise Edge [23]; SAP via AppFlow [54]; Siemens Industrial Edge hosting [23].	OPC UA connector [25]; Power BI / Microsoft 365 native [22]; SAP via Fabric mirroring (preview) [55]. *Assessment:* most Indian Tier-1 plants already run Microsoft 365 and Entra ID, so identity, Teams alerts and Excel exports come for free.	OPC UA via MC/Litmus [31]; strongest SAP-to-analytics path through the BigQuery Connector for SAP (SLT, no licence charge) [56].
Ease of use	*Assessment:* mature consoles, but the IIoT stack is now a set of parts (Grafana + Kinesis + Athena + Lambda) after the retirements [7][8][13].	*Assessment:* Fabric is SaaS and the most turnkey for Power BI users [22]; IoT Operations requires Kubernetes and Arc skills [20], which a plant IT team must learn or buy.	*Assessment:* BigQuery is the simplest analytics service, but the factory must own the broker or buy the edge [10][31].
Indicative cost (INR)	Per-message pricing [35]: ≈ **₹43,100/month** for the edge-batched design (S2). Naive one-message-per-reading costs ≈ ₹30 lakh/month, so edge batching is essential (`cost_model.md`).	Prepaid IoT Hub units make the bill predictable but dominant: ≈ **₹80,100/month** (S2), of which ₹66,300 is IoT Hub. A variant that bypasses IoT Hub is ≈ ₹13,800 (not yet validated).	≈ **₹32,900/month** (S2) using per-TiB Pub/Sub [38], *excluding* the Litmus edge licence and assuming self-run broker VMs. India list prices not retrievable.

B.4 Plant-specific requirements

Requirement	AWS	Azure	GCP
OPC UA to Fanuc + Siemens at Chakan, Oragadam, Manesar	SiteWise Edge OPC UA [23]; FOCAS via partners [15]	IoT Operations OPC UA connector [25]; FASOPC / SINUMERIK OPC UA licences [34][33]	MC / Litmus, 270+ protocols [31]
Store-and-forward during outages (frequent on Indian industrial broadband)	**Strongest documented**: SiteWise Edge keeps collecting and syncs later [23]	IoT Edge edgeHub queues with configurable TTL [57]; AIO ≤72 h, buffer not durable by default [29][26]. *Mitigation:* our gateway outbox (MQTT demo: 0 messages lost)	Depends on the Litmus edge [31]
India data residency	Mumbai hosts IoT Core, SiteWise, Greengrass [58, 59]	IoT Hub data stays in geography [27]; caveat: AIO Arc control-plane regions and Defender for IoT cloud mode are outside India [20][52]	All data services available in both India regions, storage pinnable [41, 53]

Requirement	AWS	Azure	GCP
Region next to the plants	Mumbai (Hyderabad lacks SiteWise / Greengrass) [58] [60]	Central India = Pune (P1), South India = Chennai (P2) [61]	Mumbai and Delhi (P3) [62]

B.5 Summary of the comparison

- **Azure** has the most complete current shop-floor-to-dashboard chain: the OPC UA connector, Eventstreams/Eventhouse and Power BI. It also fits the Microsoft tooling the plants already use. Its weak points are the Kubernetes skills IoT Operations needs, the 72-h offline limit, and two residency caveats.
- **AWS** is strongest on documented edge buffering and Mumbai availability. After four retirements, though, it asks the team to assemble the IIoT stack.
- **GCP** has the best analytics engine (BigQuery) and the cleanest SAP path. It has no first-party IIoT layer, so the plants' connectivity would depend on a partner.

These observations become the scores in the Part C weighted decision matrix.

5. Part C: Weighted decision matrix and recommendation

C.1 Criteria and weights (total = 100 %)

The weights reflect this scenario: brownfield CNCs, three plants on Indian industrial broadband, a small OT team, and money spent in INR.

#	Criterion	Weight	Why this weight
1	Industrial IoT & OT connectivity	**15**	Without reliable OPC UA from Fanuc and Siemens controllers, nothing downstream works. This is the first risk in a brownfield plant.
2	Edge computing & offline operation	**12**	The 2 kHz vibration must be reduced at the edge (143×, Part A), and plants lose internet.
3	Security, compliance & India residency	**12**	OT networks must be segmented (IEC 62443); automotive OEM customers expect data to stay in India.
4	Indicative cost (INR)	**13**	It matters, but cloud cost is small next to the benefit (cost model §5).
5	Time-series storage & data lake	**10**	A year of history across 100 machines, cheap cold storage, fast recent queries.
6	Stream processing & analytics	**10**	Near-real-time OEE and alarms, fleet benchmarking.
7	AI/ML	**10**	Predictive maintenance and anomaly detection (ML results).
8	Integration (MES/ERP/SAP, Microsoft 365, OPC UA)	**8**	Production orders, quality records, plant managers' tools.
9	Scalability	**5**	Growth to 1,000 machines; all three clouds handle it.
10	Ease of use & skills availability	**5**	The team that will run the platform.

C.2 Scores (1–5) with evidence

Criterion	W	AWS	Azure	GCP	Evidence (AWS / Azure / GCP)
Industrial IoT & OT connectivity	15	4	**5**	2	SiteWise Edge reads OPC UA, but IoT Events is retired and SiteWise Monitor is in maintenance [8][13][23] / IoT Operations OPC UA connector with discovery, certificates and write-back [25] / no first-party IIoT since IoT Core; partner MC needed [5][10][31]
Edge computing & offline	12	4	4	2	Greengrass V2 + SiteWise Edge with documented outage buffering; processing pack in maintenance [14][23] / IoT Operations and IoT Edge 1.6 LTS; needs Kubernetes; 72 h offline limit [20][29][19] / partner edge; GDC India availability unknown [31][49]
Security, compliance & residency	12	4	4	3	All IoT services in Mumbai [58, 59]; no IEC 62443 product certification found [51] / OT PKI, Defender for IoT; Arc control plane and Defender cloud mode outside India [20][29] [52] / storage pinning to India regions; edge security delegated to partner [31][53]

Criterion	W	AWS	Azure	GCP	Evidence (AWS / Azure / GCP)
Indicative cost (INR)	13	4	2	**5**	Computed from scenario S2: score = 5 × (cheapest ÷ this cloud), rounded to 0.5. S2 costs ₹43,121 / ₹80,085 / ₹32,928 per month
Time-series storage & data lake	10	4	**5**	4	S3 + Timestream for InfluxDB (LiveAnalytics closed) [17][18] / ADLS + Eventhouse/ADX native time series [22][28] / BigQuery/Bigtable, strong but not time-series native [32]
Stream processing & analytics	10	3	**5**	4	Kinesis + Athena assembled after IoT Analytics retired [12] / Eventstreams no-code windows + KQL + Power BI [22][28] / Dataflow + BigQuery, strongest analytics, code-first [43][48]
AI/ML	10	**5**	4	**5**	SageMaker / Azure ML, Fabric, KQL anomaly detection [22][28] / Vertex AI + BigQuery ML
Integration	8	3	**5**	4	SAP via AppFlow; Siemens Industrial Edge [23][54] / Power BI, M365, Entra native; OPC UA; SAP mirroring (preview) [25][22][55] / best SAP connector; OPC UA via partner [31][56]
Scalability	5	5	5	5	All scale well past 1,000 machines [35][28][37][10]
Ease of use & skills	5	3	**4**	3	IIoT stack now assembled from parts / Fabric SaaS, common Microsoft skills (Kubernetes needed for IoT Operations) / own broker or partner edge
Weighted total (out of 5)	**100**	**3.92**	**4.22**	**3.57**	

C.3 Sensitivity: is the winner robust?

Test	Result
Each weight shifted by ±5 points, one criterion at a time (20 runs)	**Azure wins all 20**
Equal weights (10 % each)	Azure 4.30, AWS 3.90, GCP 3.70
Capability-heavy (cost 3 %)	Azure 4.47, AWS 3.91, GCP 3.41
OT / edge-first (IoT 25, edge 20)	Azure 4.25, AWS 3.93, GCP 3.22
Analytics / AI-first (AI/ML 25, analytics 20)	Azure 4.22, GCP 3.98, AWS 3.97
Cost-heavy (cost 35 %)	**AWS 3.93**, GCP 3.83, Azure 3.82: **the winner changes**
Monte Carlo, 10,000 runs (weights ~ Dirichlet around the baseline, every score ± 0.5)	**Azure wins 88.7 %**, AWS 10.9 %, GCP 0.4 %; median Azure margin +0.26

Reading. Azure's lead comes from connectivity, analytics and integration, not from one inflated score. It is lost only when cost dominates the decision. Even then the three clouds are within 0.11 points, and the Azure IoT Hub cost could be cut further (cost model note 4).

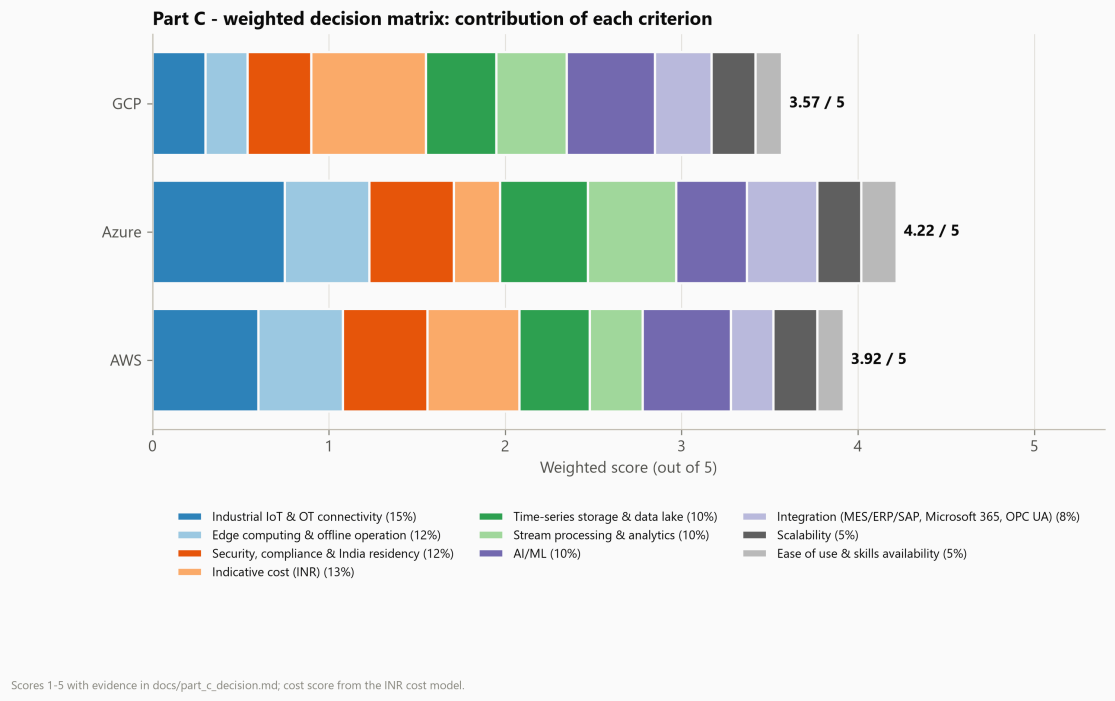


Figure 9: Weighted decision matrix: contribution of each criterion to each cloud's total

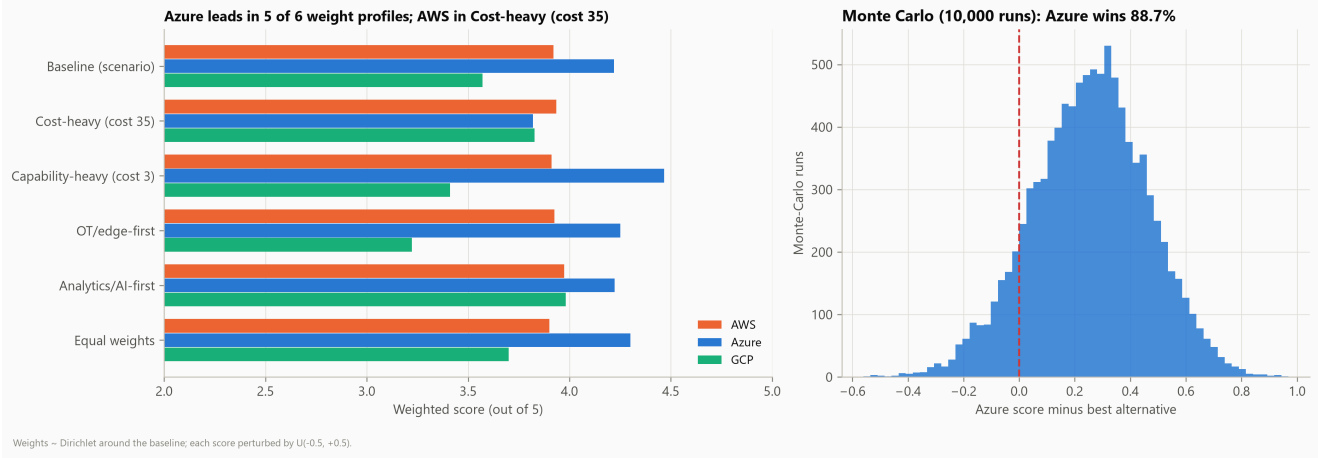


Figure 10: Sensitivity: alternative weight profiles (left) and 10,000-run Monte-Carlo distribution of Azure's margin (right)

5.1 Indicative INR cost model

Assumptions

Item	Value	Basis
Exchange rate	**1 USD = ₹94.7145**	RBI/FBIL reference rate, 21 Oct 2026, 1:00 pm IST [63]
Month / year	30 days (720 h) / 365 days	assignment convention
Fleet	100 machines, 24×7	scenario
Regions	AWS ap-south-1 (Mumbai), Azure Central India (Pune), GCP asia-south1 (Mumbai)	GCP India prices not retrievable, so default-region list prices are used [38, 47]
Message pattern	1 MQTT message per machine per second (edge batching), except S0	Part A
Payload sizes	measured JSON: 10 sensors at 1 Hz = 296 B; 100 samples at 100 Hz = 729 B; one reading = 125 B; 42 edge features ≈ 564 B; 2 kHz × 3 axes ≈ 42 KB	Part A (`part_a_mqtt_payloads.json`)

Item	Value	Basis
Storage	12-month rolling retention in the data lake; Parquet+Zstd compression 6.9× (measured on real UMich signals), 5× for 2 kHz raw (measured on Bosch)	Part A
AWS	IoT Core $1.05 per million (tiered), metered in 5 KB units, plus 1 rule and 1 action per message; Kinesis 2 shards; S3 Standard; RDS PostgreSQL db.t4g.medium (Oregon price, Mumbai not retrievable)	[24][35][39][36][45]
Azure	IoT Hub cheapest tier mix (S1 400k/day $35; S2 6M/day $350; S3 300M/day $3,500; 4 KB metering); Event Hubs 2 TU; ADLS Gen2 Hot; PostgreSQL Flexible B2s	[37]
GCP	Pub/Sub $40/TiB plus 3 self-managed MQTT broker VMs (assumption: $60/VM-month); Dataflow 1 worker (**assumption**, $0.07/h); Cloud Storage; Cloud SQL 2 vCPU / 4 GiB. The Manufacturing Connect (Litmus) licence is **not included**: no public price.	[30][38, 47]

Scenarios

	Data sent to the cloud per machine-second	Messages / month (fleet)
S0	raw readings, **one MQTT message per reading** (10 × 1 Hz + 100 × 100 Hz)	28.5 billion
S1	raw 1 Hz (10 sensors) + raw 100 Hz vibration, batched into 1 message (1,025 B)	259.2 million
S2	raw 1 Hz + **42 edge vibration features** (860 B), batched	259.2 million
S3	raw 1 Hz + raw **2 kHz tri-axial** accelerometer (42 KB), batched	259.2 million

Results (₹ per month)

Scenario	AWS	Azure	GCP
S0 one message per reading	₹29,91,825	₹14,25,721	₹55,447
S1 raw 1 Hz + raw 100 Hz	₹43,297	₹80,226	₹33,216
S2 raw 1 Hz + edge features (recommended)	**₹43,121**	**₹80,085**	**₹32,928**
S3 raw 2 kHz to the cloud	₹2,84,788	₹3,94,487	₹1,19,208

Component breakdown: `docs/results/part_c_cost_tables.md`. Figure: `docs/figures/part_c_cost_model.png`.

What the numbers say

1. **Edge batching is the largest cost lever.**
   - Sending each reading as its own MQTT message (S0) costs ₹14–30 lakh/month on the per-message-priced clouds.
   - Batching one message per machine-second at the gateway (S1) brings that down to ₹0.4–0.8 lakh, **35–70× cheaper**.
2. **At 100 Hz, edge features save bandwidth, not much money.**
   - S1 and S2 cost almost the same, because both payloads fit inside one 4 KB (Azure) or 5 KB (AWS) billing unit and storage is small (~0.5 TB compressed per year).
   - The case for edge processing at 100 Hz rests on latency, outage resilience and data quality (Part G), not on rupees.
3. **At the real sensor rate, edge features save ₹0.9–3.1 lakh per month.**
   - The Bosch accelerometers sample at 2 kHz on 3 axes. Streaming that raw (S3) costs ₹1.2–3.9 lakh/month.
   - Edge features (S2) save **₹2.42 lakh/month on AWS, ₹3.14 lakh on Azure and ₹0.86 lakh on GCP** (₹29–38 lakh/year on AWS/Azure).
   - The 143× reduction measured in Part A is what makes this possible.
4. **Platform ranking for S2:** GCP < AWS < Azure. But:

- **GCP** excludes the Litmus edge licence that its manufacturing path requires, and relies on assumed broker and Dataflow costs.
- **Azure**'s bill is dominated by two prepaid IoT Hub S2 units (₹66,300). *Design note (not scored):* Fabric Eventstreams accept MQTT sources directly [22], and IoT Operations has its own MQTT broker and data flows at the edge [26]. A path that bypasses IoT Hub is therefore plausible. In that variant Azure S2 would total about ₹13,800/month (Event Hubs + storage + PostgreSQL). It must be validated in a proof-of-concept before it is relied on, so it is **not** used in the scored cost.

5. **The spread between clouds is small in absolute terms.** At ₹33k–80k/month (₹4–10 lakh/year), cloud cost is about 0.7–1.6% of the ₹5.8 crore/year indicative benefit (ML business-impact estimate). Cost is weighted 13%, not more, for that reason.

One-time edge costs (same for every cloud, assumptions)

Item	Qty	Indicative ₹
Industrial PC gateway (fanless, 8-core, 32 GB, 1 TB SSD for store-and-forward), 2 per plant for HA	6	9,00,000
OPC UA enablement on controllers (FASOPC / SINUMERIK OPC UA licence, or edge FOCAS/S7 driver)	100	30,00,000
Vibration sensors + DAQ for spindle monitoring (one tri-axial per machine)	100	35,00,000
Industrial firewall / DMZ switch per plant (IEC 62443 zone boundary)	3	6,00,000
Total one-time		**≈ ₹80 lakh**

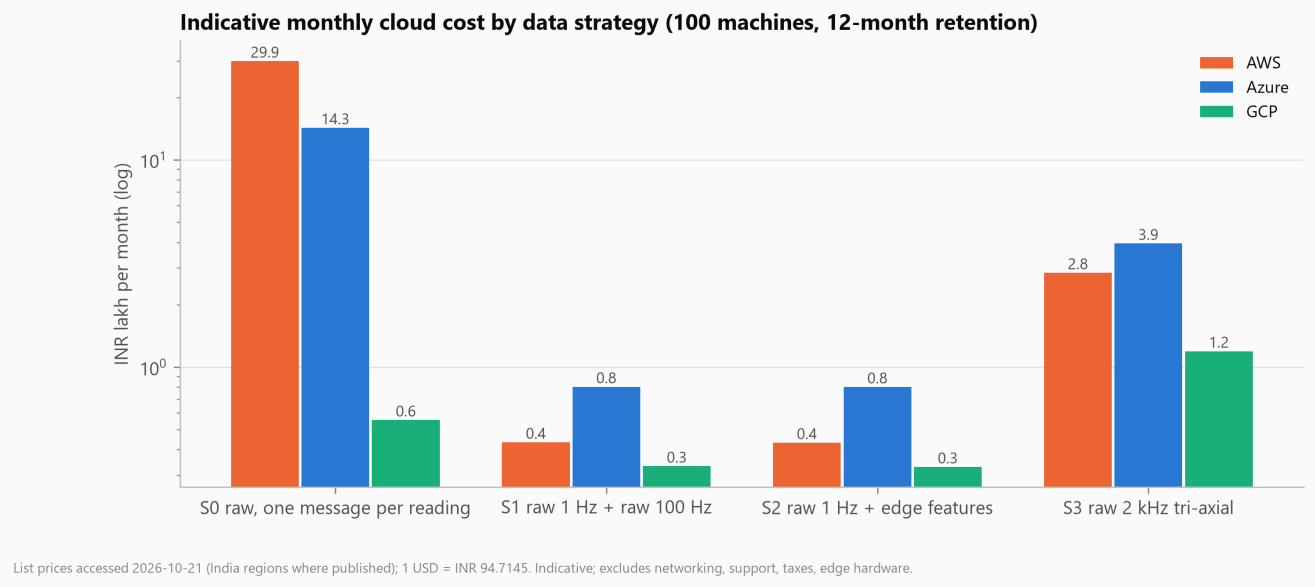


Figure 11: Indicative monthly cloud cost by data strategy (log scale)

C.4 Final recommendation: Microsoft Azure (Central India), with an edge-first design

The scenario, not just the 4.22, supports this:

1. **Brownfield OPC UA is the critical path, and Azure covers it best.**
   - IoT Operations' OPC UA connector gives asset discovery, certificate management and write-back [25] for the Fanuc FASOPC and SINUMERIK OPC UA servers [34][33].
   - That turns 100 controllers into registered assets without per-machine custom code.
   - AWS SiteWise Edge is close, but alarm rules and dashboards would have to be rebuilt after the IoT Events and Monitor retirements.
   - GCP would put the whole shop-floor layer on a partner contract.
2. **The plants already run on Microsoft.**
   - Typical Indian Tier-1 suppliers use Entra ID, Microsoft 365 and Excel/Power BI.

- Fabric's Real-Time Dashboards and Power BI reach plant managers, quality engineers and the OEE review meeting without new tools or new logins [22].

3. **Data stays near the plants.**
   - Central India is in Pune, beside Chakan, with South India (Chennai, beside Oragadam) as its pair [61]. IoT Hub keeps data in the geography [27].
   - **Mitigations for the two residency caveats:**
     - The IoT Operations Arc control-plane region is outside India [20]. Only resource metadata goes there, and telemetry sinks are configured to Central India.
     - Defender for IoT's cloud mode routes to East US [52]. Use its air-gapped / locally managed sensor mode.
4. **Edge-first removes Azure's two weaknesses.**
   - **72 h offline limit:** the plant gateway runs its own durable store-and-forward outbox, demonstrated with 0 messages lost in the MQTT demo, sized for 7 days. A 7-day buffer for the 40 machines at Chakan is about 21 GB (860 B/s per machine), which fits on the gateway's 1 TB SSD.
   - **Cost:** batching per machine-second and edge features keep the bill at about **₹0.8 lakh/month** (about ₹0.14 lakh if IoT Hub can be bypassed, cost model note 4). That is about 1.6% of the indicative ₹5.8 crore/year benefit.
5. **Skills are the main risk, and it is manageable.**
   - IoT Operations needs a small Kubernetes footprint (K3s / AKS Edge Essentials) at each plant [20].
   - Mitigation: start with one pilot line at Chakan (P1, 40 HMC/VMC machines), use a system integrator for the first cluster, and train two plant IT staff before rolling out to P2 and P3.

When to choose differently.

- **AWS:** if management makes monthly cloud cost the dominant criterion (≥ 35 % weight), or if the plants standardise on Siemens Industrial Edge, where SiteWise Edge can be hosted directly [23].
- **GCP:** for a data-science-led programme with heavy SAP analytics, *and* if a Litmus Edge contract is acceptable.

6. Part D: PostgreSQL database design, queries and results

Platform and layout.

- The database runs on PostgreSQL 16 (any 16/17 works).
- Three schemas separate the concerns:
  - `raw`: the four source datasets as published (921,172 rows);
  - `core`: master data and the hourly fact table;
  - `analytics`: views and materialized views for the dashboard.
- The full DDL is in `sql/01_schema.sql`.

Design choices.

- **Keys:** primary key `(machine_id, ts)`, which also serves as the main access index.
- **Constraints:** foreign keys to the plant, machine, part-family and alarm masters; CHECK constraints for non-negative counts, `reject_count ≤ production_count`, running minutes 0–60, and alarm-flag consistency.
- **Status:** an ENUM `machine_status_t` (RUNNING / IDLE / ALARM / MAINTENANCE / OFF).
- **Additional indexes:** `ts`, `(plant_id, ts)`, and a partial index on alarm rows (0.5% of rows).
- **Time zone:** the database is set to IST.
- **Required column names:** the view `analytics.v_machine_data_required` exposes `Machine_ID`, `Timestamp`, `Temperature`, `Vibration`, `RPM`, `Current`, `Production_Count`, `Machine_Status`, `Alarm` and `Reject_Count`.

- **Production path:** a TimescaleDB setup (hypertable, continuous aggregates, compression) is documented, commented out, in the schema.

Loading. COPY loaded the 876,100 fact rows at about 47,000 rows/s. The table takes 162 MB including indexes, about 185 bytes per row, compared with about 20 bytes per row in Parquet+Zstd.

Part D - Entity-relationship diagram (generated from the PostgreSQL catalog)

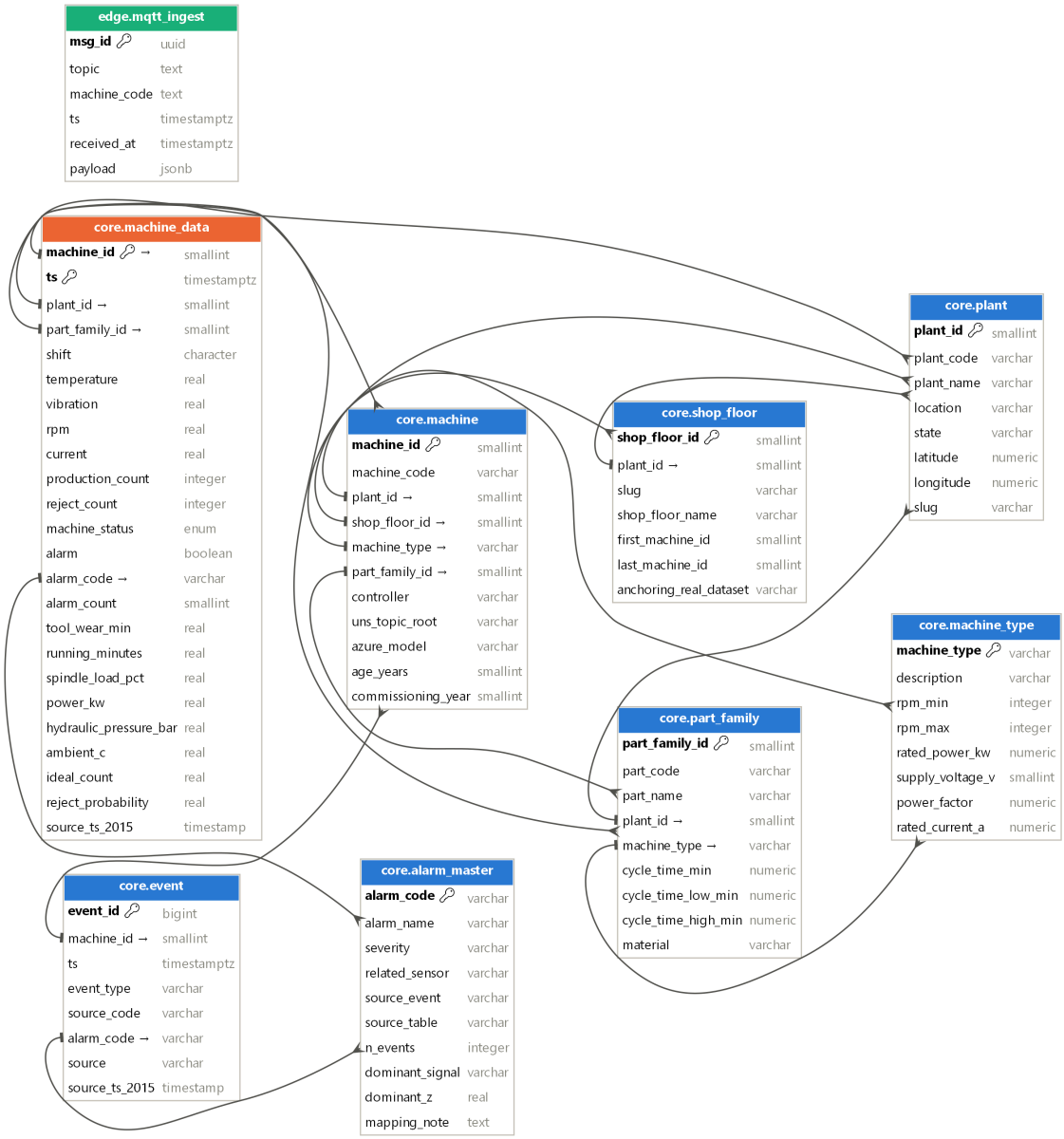


Figure 12: Entity-relationship diagram generated from the PostgreSQL catalog (orange: fact table; green: MQTT ingest table)

```sql
CREATE TABLE core.machine_data ( machine_id SMALLINT NOT NULL REFERENCES core.machine, ts TIMESTAMPTZ NOT NULL, plant_id SMALLINT NOT NULL REFERENCES core.plant, part_family_id SMALLINT NOT NULL REFERENCES core.part_family, temperature REAL, vibration REAL, rpm REAL, current REAL, production_count INTEGER NOT NULL DEFAULT 0 CHECK (production_count >= 0), reject_count INTEGER NOT NULL DEFAULT 0 CHECK (reject_count >= 0), machine_status core.machine_status_t NOT NULL, alarm BOOLEAN NOT NULL DEFAULT FALSE, alarm_code VARCHAR(8) REFERENCES core.alarm_master, tool_wear_min REAL, running_minutes REAL NOT NULL CHECK (running_minutes BETWEEN 0 AND 60), ... PRIMARY KEY (machine_id, ts), CHECK (reject_count <= production_count), CHECK (alarm = (alarm_code IS NOT NULL)));
```

6.1 The eight required queries

The query logic, all from `sql/03_required_queries.sql`:

#	Question	Logic
Q1	Average temperature per machine	Average over hours with running minutes > 0
Q2	Machine with the most parts	`SUM(production_count)` with `FETCH FIRST 1 ROWS WITH TIES`
Q3	Highest average vibration	Average while running, with its ISO 10816-3 zone
Q4	Alarm count per machine	Alarm events, alarm hours, and critical (failure) hours
Q5	Parts on a selected day	The date is a parameter; the IST day uses a half-open range `[d, d+1)` so the index on `ts` stays usable
Q6	Longest operating duration	Gaps-and-islands: `ts − row_number() × 1 h` is constant within a continuous RUNNING run
Q7	Average RPM while running	RUNNING hours only
Q8	Highest rejection rate	`SUM(reject_count) / SUM(production_count)`, a ratio of sums, not an average of hourly ratios

Table 3: Q1 Average temperature per machine (running hours), highest first (first 5 of 100 rows). Source: `sql/results/q1_avg_temperature_per_machine.csv`.

machine_id	machine_code	plant_code	avg_temperature_c	max_temperature_c	running_hours
57	P2-HOB-07	P2	39.13	45.24	8550
59	P2-HOB-09	P2	39.13	45.07	8553
64	P2-TURN-14	P2	39.13	45.21	8538
43	P2-TURN-03	P2	39.12	45.13	8528
48	P2-TURN-08	P2	39.12	45.09	8545

Table 4: Q2 Machine that produced the most parts. Source: `sql/results/q2_machine_most_parts.csv`.

machine_id	machine_code	part_name	cycle_time_min	total_parts
80	P3-TURN-05	Brake disc	2	199677

Table 5: Q3 Highest average vibration while running (first 5 of 10 rows). Source: `sql/results/q3_highest_avg_vibration.csv`.

rnk	machine_id	machine_code	machine_type	avg_vibration_mm_s	max_vibration_mm_s	iso_zone_of_average
1	37	P1-VMC-13	VMC	1.856	4.741	B (acceptable)
2	83	P3-TURN-08	TURN	1.851	4.882	B (acceptable)
3	64	P2-TURN-14	TURN	1.85	5.142	B (acceptable)
4	49	P2-TURN-09	TURN	1.848	5.128	B (acceptable)
5	1	P1-HMC-01	HMC	1.844	5.078	B (acceptable)

Table 6: Q4 Alarm count per machine, most first (first 5 of 100 rows). Source: `sql/results/q4_alarm_count_per_machine.csv`.

machine_id	machine_code	alarm_events	alarm_hours	critical_alarm_hours
22	P1-HMC-22	75	67	13
99	P3-TURN-16	73	67	19
17	P1-HMC-17	64	56	13
49	P2-TURN-09	63	58	11
78	P3-TURN-03	63	61	9

Table 7: Q5 Parts produced on the selected day (16 Jun 2025, IST). Source: `sql/results/q5_parts_on_selected_day.csv`.

plant	parts_produced	parts_rejected	selected_day
ALL	23974	381	2025-06-16
P1	4172	52	2025-06-16
P2	10652	173	2025-06-16
P3	9150	156	2025-06-16

Table 8: Q6 Longest continuous operating duration (first 5 of 10 rows). Source: `sql/results/q6_longest_operating_duration.csv`.

rnk	machine_code	run_start	run_end	longest_uninterrupted_running_hours	days	total_running_hours
1	P1-HMC-18	2025-12-01 05:00:00+05:30	2025-12-09 22:00:00+05:30	209	8.7	6,771.9
2	P1-VMC-01	2025-11-17 07:00:00+05:30	2025-11-25 07:00:00+05:30	192	8	6,785.5
3	P1-VMC-04	2025-01-12 07:00:00+05:30	2025-01-20 00:00:00+05:30	185	7.7	6,804.1
4	P2-HOB-09	2025-07-25 07:00:00+05:30	2025-08-01 04:00:00+05:30	165	6.9	6,798.8
5	P3-TURN-15	2025-01-04 19:00:00+05:30	2025-01-11 15:00:00+05:30	164	6.8	6,744.5

Table 9: Q7 Average RPM while running (first 5 of 100 rows). Source: `sql/results/q7_avg_rpm_while_running.csv`.

machine_id	machine_code	machine_type	avg_rpm_running	rpm_min	rpm_max
1	P1-HMC-01	HMC	5325	2000	8000
2	P1-HMC-02	HMC	5330	2000	8000
3	P1-HMC-03	HMC	5340	2000	8000
4	P1-HMC-04	HMC	5326	2000	8000
5	P1-HMC-05	HMC	5337	2000	8000

Table 10: Q8 Highest rejection rate (first 5 of 10 rows). Source: `sql/results/q8_highest_rejection_rate.csv`.

rnk	machine_code	part_name	produced	rejected	rejection_rate_pct
1	P2-GRIND-06	Gear shaft (finish grind)	157438	3106	1.973
2	P2-GRIND-01	Gear shaft (finish grind)	158565	3013	1.9
3	P2-TURN-06	Gear shaft	113160	2131	1.883
4	P2-GRIND-04	Gear shaft (finish grind)	157903	2954	1.871
5	P2-GRIND-10	Gear shaft (finish grind)	157990	2954	1.87

Interpretation.

- **Q2 and Q8 follow cycle time.** Brake-disc turning centres (2-min cycle) produce the most parts. Finish grinders, which run at high utilisation with tool wear, have the highest reject rates, about 1.9–2.0%.
- **Q3 sits in ISO zone B.** Fleet-average vibration is in zone B (1.4–2.8 mm/s), so the useful signal is in the excursions, not the averages (query A7 below).
- **Q6 shows one exceptionally long run.** P1-HMC-18 ran 209 h (8.7 days) without interruption.

6.2 Advanced queries (bonus)

`sql/04_advanced_queries.sql` adds:

- OEE (availability × performance × quality) per machine and per plant-month;

- MTBF and MTTR, from failure events and the maintenance islands that follow them;
- shift-wise production;
- an alarm Pareto with cumulative share;
- daily trends with 7-day moving averages, `LAG` and `RANK`;
- vibration anomalies against a rolling 168-hour per-machine baseline, linked to failures in the following 48 h;
- reject rate by tool-wear band;
- dashboard materialized views.

Table 11: A1 OEE per machine, lowest first (first 5 of 100 rows). Source: `sql/results/a1_oee_per_machine.csv`.

machine_code	plant_code	machine_type	availability	performance	quality	oee
P2-GRIND-06	P2	GRIND	0.782	0.971	0.98	0.744
P3-TURN-03	P3	TURN	0.78	0.972	0.983	0.745
P1-HMC-13	P1	HMC	0.783	0.97	0.982	0.746
P1-VMC-13	P1	VMC	0.781	0.971	0.983	0.746
P1-VMC-14	P1	VMC	0.783	0.97	0.983	0.747

Table 12: A3 MTBF and MTTR, least reliable first (first 5 of 100 rows). Source: `sql/results/a3_mtbf_mttr_per_machine.csv`.

machine_code	plant_code	failures	running_hours	mtbf_hours	mttr_hours	inherent_availability
P3-TURN-16	P3	18	6713	373	7.3	0.981
P3-TURN-15	P3	16	6745	422	8.3	0.981
P1-HMC-22	P1	14	6750	482	8.4	0.983
P1-VMC-13	P1	14	6763	483	8.1	0.984
P1-HMC-17	P1	14	6765	483	8.1	0.983

Table 13: A5 Alarm Pareto with cumulative share. Source: `sql/results/a5_alarm_pareto.csv`.

alarm_code	alarm_name	severity	events	pct	cumulative_pct
AL-1010	Spindle drive overcurrent / overload	WARNING	1010	21.58	21.58
AL-2010	Spindle speed deviation (actual < command)	WARNING	988	21.11	42.69
AL-2020	Axis following error (servo lag)	WARNING	838	17.91	60.6
AL-3010	Hydraulic pressure abnormal (tool clamp)	WARNING	727	15.53	76.13
AL-4010	High spindle vibration	WARNING	356	7.61	83.74
AL-9200	Spindle gearbox / belt failure	CRITICAL	259	5.53	89.27
AL-9100	Spindle motor / drive failure	CRITICAL	192	4.1	93.38
AL-9400	Tool breakage / spindle bearing damage	CRITICAL	179	3.82	97.2
AL-9300	Hydraulic unit / tool-clamp failure	CRITICAL	131	2.8	100

Table 14: A7 Vibration anomalies (z > 3 vs rolling 7-day baseline) and how often a failure followed within 48 h (first 5 of 100 rows). Source: `sql/results/a7_vibration_anomalies.csv`.

machine_code	anomalies	followed_by_failure_48h	precision_pct	max_z	max_vibration_mm_s
P1-VMC-13	41	20	48.8	11.71	4.74
P1-HMC-13	38	6	15.8	4.43	5.42
P1-HMC-01	36	16	44.4	4.1	5.08
P2-HOB-01	36	7	19.4	4.88	5.5
P3-VMC-01	36	12	33.3	49.92	4.6

Table 15: A8 Reject rate by tool-wear band. Source: `sql/results/a8_tool_wear_vs_quality.csv`.

wear_band	wear_range_min	parts	rejects	reject_rate_pct
1	0-30	1259471	6754	0.536
2	30-60	1367410	9749	0.713
3	60-90	1313398	12708	0.968
4	90-120	1288572	16367	1.27
5	120-150	1263957	21804	1.725
6	150-180	1232740	28137	2.282
7	180-210	929190	27023	2.908
8	210-240	294194	11009	3.742
9	240-246	8881	415	4.673

7. Part F: Industrial cloud architecture on Azure



Figure 13: Required linear architecture with each block mapped to an Azure service (middle) and to what this project built (right)

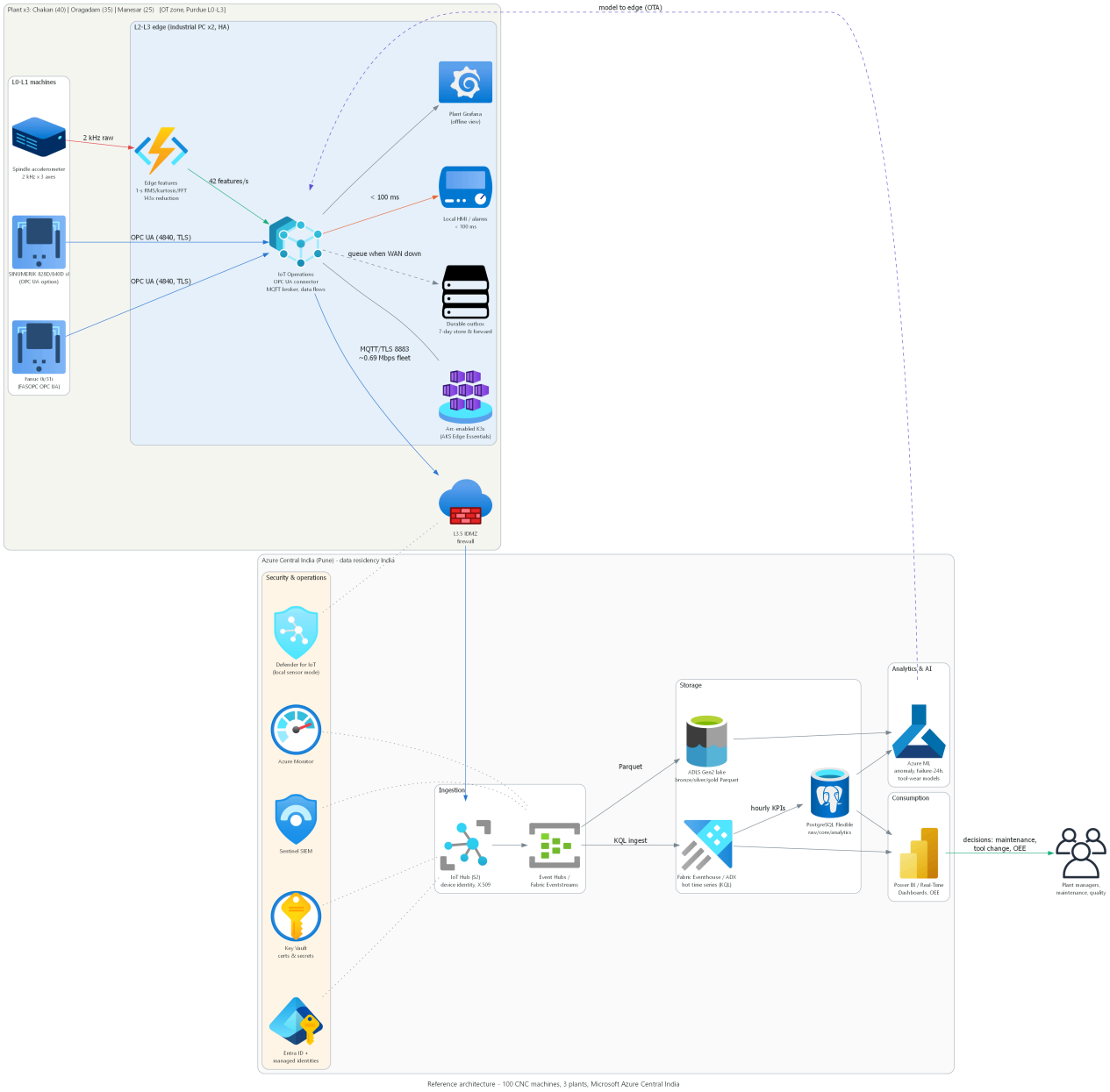


Figure 14: Azure reference architecture: plant edge (IoT Operations on Arc-enabled K3s) to Azure Central India

F.1 Block-to-service mapping

#	Block	Azure service / component	Role in this factory	Evidence / status
1	**CNC / PLC / Sensors**	Fanuc 0i-MF / 31i-B5 and SINUMERIK 828D / 840D sl controllers; one tri-axial spindle accelerometer per machine (2 kHz)	10 slow streams (spindle temperature, coolant temperature, spindle vibration RMS, RPM, spindle current, feed-axis load, hydraulic pressure, coolant flow, power, door/cycle status) plus 1 fast vibration channel	Controller OPC UA needs FASOPC or the SINUMERIK OPC UA option [34] [33]
2	**OPC UA / MQTT**	OPC UA (port 4840, Sign & Encrypt) machine → gateway; MQTT 3.1.1/5 over TLS gateway → cloud	Unified Namespace topics: `tier1/<site>/<area>/<machine>/<stream>` (e.g. `tier1/chakan/powertrain/hmc-07/vibration/features`)	Event Grid MQTT broker GA with MQTT v5 [21]

#	Block	Azure service / component	Role in this factory	Evidence / status
3	**Edge Gateway**	**Azure IoT Operations** on **Arc-enabled K3s / AKS Edge Essentials**, two industrial PCs per plant (HA). OPC UA connector, MQTT broker, data flows. A durable outbox sized for 7 days.	Protocol conversion; 1-s vibration features (143× reduction); per-second batching; local alarms; store-and-forward	AIO GA 1.2–1.4 [20]; OPC UA connector [25]; AIO offline ≤ 72 h, so a durable outbox is added [29][26]
4	**Cloud Data Ingestion**	**Azure IoT Hub (S2 ×2)**: per-gateway X.509 identity, routing. **Event Hubs** for fan-out; **Fabric Eventstreams** as the stream-processing front end	259.2 M batched messages per month (8.64 M/day)	IoT Hub tiers [37]; Central India pricing []; Eventstreams MQTT source [22]
5	**Data Lake / Time-Series Storage**	**ADLS Gen2** (bronze raw JSON → silver Parquet → gold aggregates); **Fabric Eventhouse / Azure Data Explorer** for the hot 30–90 days with KQL time-series functions	12-month lake retention ≈ 0.4 TB compressed (6.9× Parquet+Zstd measured)	ADLS pricing []; ADX time-series [28]; Time Series Insights retired, replaced by ADX [6]
6	**SQL Database**	**Azure Database for PostgreSQL Flexible Server**, with the schema in `sql/01_schema.sql` unchanged (raw / core / analytics; TimescaleDB path documented)	Hourly machine KPIs, master data, alarms, OEE / MTBF views for apps and reports	Flexible Server pricing in Central India []
7	**Python Analytics / AI**	**Azure Machine Learning** (training, registry, batch scoring) and Fabric notebooks; models deployed to the edge as containers through IoT Operations	Isolation Forest vibration anomaly (ROC-AUC 0.98), 24-h failure risk, failure-mode classifier	Eventhouse anomaly functions [22] [28]
8	**Dashboard**	**Power BI** and **Fabric Real-Time Dashboards** for management; this project's **Streamlit** app (App Service or Streamlit Community Cloud)	OEE, quality, alarms, machine health, PdM risk	Power BI native in Fabric [22]
9	**Manufacturing Decision**	**Fabric Activator** → Teams / e-mail alerts; maintenance work orders in the CMMS / SAP PM	Wear-based tool change (reject rate 1.50% → 0.92%, indicative); risk-ranked maintenance	Activator in Fabric RTI [22]
–	**Security & operations**	Entra ID + managed identities, Key Vault (certificates), **Defender for IoT** OT sensor in *locally managed* mode, Sentinel, Azure Monitor	OT network visibility, secrets, audit	AIO PKI and X.509 [29]; Defender for IoT cloud mode routes to East US, so local mode is used [52]

Edge hardware per plant (one-time cost, `cost_model.md`):

- 2 × fanless industrial PCs (8 cores, 32 GB RAM, 1 TB SSD) running K3s, in a high-availability pair;
- an industrial firewall at the Level 3.5 DMZ;
- a SPAN port for the Defender for IoT sensor.

F.2 Why this shape

- **Thin cloud ingress, rich edge.**
  - Part A measured 19.2 Mbps of raw vibration for the fleet against 0.13 Mbps of features. Only features and 1 Hz data cross the WAN: about 0.69 Mbps for the whole fleet, under 1% of a typical 100 Mbps plant link.
  - The naive one-message-per-reading design would cost ₹14–30 lakh/month in ingestion (`cost_model.md`).
- **Two time-series tiers.**
  - Eventhouse/ADX for fast recent queries (the dashboard's "last 30 days").
  - ADLS Parquet for cheap long history and model training.
  - PostgreSQL holds only hourly KPIs and master data: 185 B/row measured, so 876,100 rows take 162 MB.
- **Region choice.**
  - Central India is in Pune, beside the Chakan plant, and is paired with South India in Chennai, beside Oragadam [61].
  - IoT Hub keeps data in the geography [27].
  - The two residency caveats and their mitigations are in Part C §C.4.

F.3 Security zoning (Purdue / IEC 62443), diagram `03_security_zones`

Zone (Purdue level)	Contents	Target security level (SL-T)	Conduit out
Cell / machine zone (L0–L2), one VLAN per shop floor	CNC controllers, PLCs, safety I/O, accelerometers, HMIs	SL 2–3	**C1**: OPC UA 4840 Sign & Encrypt to the gateway only
Site operations zone (L3)	Edge gateway cluster, plant historian / Grafana, Defender for IoT sensor (passive SPAN)	SL 3	**C2**: MQTT/TLS 8883, outbound only
Industrial DMZ (L3.5)	MQTT relay, patch server, jump host for vendor access (MFA, session recording)	SL 3	**C3**: TLS 1.2+ to Azure via VPN / private peering
Enterprise IT (L4)	ERP/SAP, MES reports, office users	SL 2	**C4**: HTTPS with Entra ID
Cloud (L5)	IoT Hub, ADX, ADLS, PostgreSQL (private endpoints), Azure ML, Power BI	SL 2	–

Rules:

- All flows are initiated from the lower level to the higher one. There is no inbound path from the internet or Level 4 to Levels 0–2.
- Firewalls deny by default at every conduit.
- Machine protection and alarms never depend on the cloud.
- Every gateway has its own X.509 certificate, kept in Key Vault and rotated.
- Vendor remote access only goes through the time-boxed, audited jump host (conduit C5).

F.4 Working demonstration in this repository

Architecture block	Implemented here	Evidence
Edge feature extraction	`python/edge/bosch_features.py` on all 1,702 real Bosch files	143× reduction per machine-second; 120× measured on the full dataset (`docs/results/edge_reduction.json`)
Edge gateway + MQTT + store-and-forward	`python/edge/edge_gateway.py`, `mqtt_broker.py`, `cloud_subscriber.py`	482 produced / 482 stored / 0 lost / 0 duplicates across a subscriber outage and a 12-s WAN outage (`docs/results/mqtt_demo_summary.json`, figure `edge_mqtt_store_and_forward.png`)
SQL database	PostgreSQL 16 with `sql/01_schema.sql`	876,100 fact rows loaded at about 47k rows/s with COPY; 8 required + 10 advanced queries (`sql/results/`)
Analytics / AI	`python/ml/`	Model results in `docs/results/ml_*.json`

Architecture block	Implemented here	Evidence
Dashboard	`dashboard/` (Streamlit, 8 pages)	Local mode reads PostgreSQL; hosted mode reads Parquet

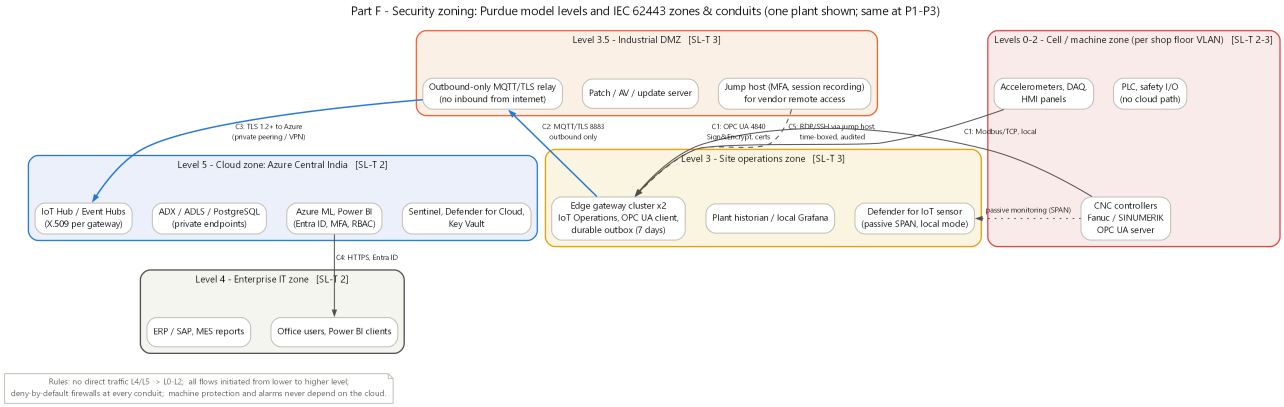


Figure 15: Security zoning: Purdue levels and IEC 62443 zones and conduits [64]

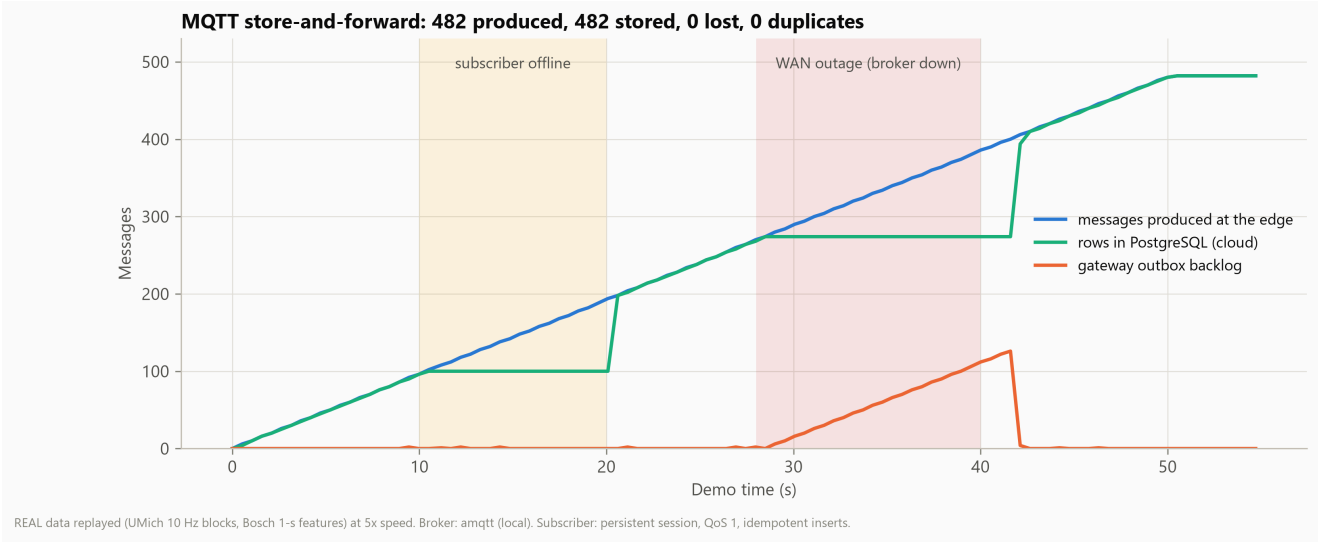


Figure 16: MQTT store-and-forward demonstration with real data: subscriber outage and WAN outage, 0 messages lost

8. Part G: Edge vs cloud decision

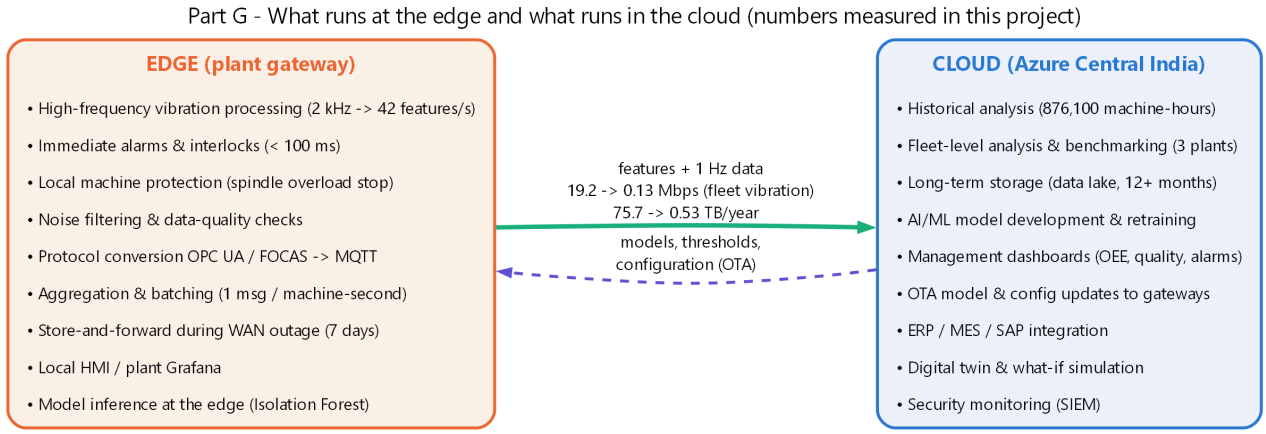


Figure 17: Edge vs cloud split with measured data rates

G.1 Measured evidence used below

Quantity	Value	Source
Raw vibration per machine (2 kHz × 3 axes, float32)	24,000 B/s	Part A, Bosch data
Edge features per machine (42 values/s)	168 B/s, a **143× reduction** (120× measured over all 24.8 h of data)	`edge_reduction.json`
Fleet uplink, raw vibration vs features	**19.2 Mbps → 0.13 Mbps**	`edge_reduction.json`
Fleet storage per year, raw vibration vs features	**75.7 TB → 0.53 TB**	`edge_reduction.json`
Cloud bill, raw 2 kHz vs edge features (Azure, per month)	₹3,94,487 → ₹80,085, a **₹3.14 lakh/month** saving	`cost_model.md`
Cloud bill, one message per reading vs edge batching (AWS / Azure)	₹29.9 lakh / ₹14.3 lakh → ₹0.43 lakh / ₹0.80 lakh per month	`cost_model.md`
Store-and-forward during outage	0 of 482 messages lost, 126-message peak backlog, 0 duplicates	`mqtt_demo_summary.json`
Good vs bad separability from edge features alone	ROC-AUC 0.98 (Isolation Forest trained on good only); 0.98 on a 2019 → 2021 time hold-out	`ml_bosch_anomaly.json`

G.2 Operation placement and justification

L = latency, R = reliability, B = bandwidth, C = cost, S = cybersecurity, Sc = scalability.

Operation	Where	Justification
High-frequency vibration processing (2 kHz → RMS, kurtosis, band energies, dominant frequency)	**Edge**	**B:** 19.2 → 0.13 Mbps for the fleet. **C:** ₹3.1 lakh/month less on Azure. **R:** works offline. The features keep the fault signal (AUC 0.98), so little is lost by not shipping raw data.
Immediate alarms and interlocks (spindle overload, high vibration)	**Edge**	**L:** must act in under 100 ms. A cloud round-trip from Chakan to Pune adds tens to hundreds of milliseconds, and the WAN can be down. **R:** a protective function must not depend on the internet.
Local machine protection (feed hold, spindle stop)	**Edge**, inside the CNC/PLC	**S:** stays in the cell zone, with no write path from the cloud (IEC 62443 conduits). **L/R** as above.
Noise filtering and data-quality checks (range, stuck sensor, timestamp sanity)	**Edge**	**B/C:** bad data never pays for ingestion or storage. **R:** quality flags are added close to the source, while context is still known.
Protocol conversion (OPC UA / FOCAS / S7 → MQTT, Unified Namespace)	**Edge**	**S:** the cloud never talks to controllers. **Sc:** every plant publishes the same topic model, so plant 4 is a configuration change.
Data aggregation and batching (one message per machine-second)	**Edge**	**C:** 35–70× lower ingestion bill than one message per reading. **B:** fewer, larger TLS packets.
Store-and-forward during WAN outage	**Edge**	**R:** 0 lost and 0 duplicates in the demo. A 7-day buffer for 40 machines is about 21 GB, sized to outlast the IoT Operations 72-h offline limit [29].
Local HMI / plant dashboard	**Edge**	**R:** operators keep their view during outages. **L:** live, sub-second refresh.
Model inference at the edge (Isolation Forest on 1-s features)	**Edge**	**L:** an anomaly flag within 1 s. **B:** only scores and flagged windows are sent. **R:** works offline.
Historical analysis (876,100 machine-hours, MTBF/MTTR, OEE trends)	**Cloud**	**Sc:** needs the whole fleet's history. **C:** cheap object storage (₹0.7k–1k/month for a year of data).
Fleet-level analysis and benchmarking (3 plants, 100 machines)	**Cloud**	Only the cloud sees all plants at once, for example OEE by plant and the alarm Pareto.
Long-term storage (data lake, ≥ 12 months, audit)	**Cloud**	**C:** ADLS ≈ ₹1.9/GB-month. **R:** geo-redundant pair (Pune ↔ Chennai). Edge disks are a buffer, not an archive.

Operation	Where	Justification
AI/ML model development and retraining	**Cloud**	Needs history, labels and GPU/CPU on demand (Azure ML). **Sc:** one model serves many machines. Example: the cross-machine Bosch test (M01 → M03) needs fleet data.
OTA model and configuration updates	**Cloud → edge**	**Sc:** one release pipeline for six gateways. **S:** signed images, pulled by the gateway (outbound connection only) [29].
Management dashboards (OEE, quality, alarms, PdM risk)	**Cloud**	Users are off the shop floor; Entra ID access control; Power BI / Streamlit.
ERP / MES / SAP integration	**Cloud**	Enterprise systems sit in Level 4/5. Integrating once in the cloud avoids three plant-level integrations.
Digital twin and what-if simulation	**Cloud**	Compute-heavy, fleet-wide, not time-critical.
Security monitoring (SIEM)	**Cloud**, with OT sensors at the edge	Defender for IoT sensors watch the OT network locally (passive); alerts are correlated centrally in Sentinel.

G.3 Rule of thumb that falls out of the numbers

1. **If it must happen in under 1 s, or must keep working without the internet, it belongs at the edge.** That covers protection, alarms, HMI and buffering.
2. **If it reduces data by more than 10× without losing the signal, do it at the edge.** Feature extraction (143×) and batching (35–70× cost) both qualify.
3. **If it needs more than one machine's or one plant's history, or a human outside the shop floor, it belongs in the cloud.** That covers training, fleet benchmarking, management dashboards and ERP.
4. **Keep the raw high-frequency data for a short time at the edge** (for example a 7-day ring buffer), and upload raw windows only on demand around detected anomalies. Engineers can still inspect a fault waveform, and the 75.7 TB/year of raw data never has to be stored.

9. Analytics and machine-learning results

Validation approach. All models use evaluation splits that prevent leakage:

- held-out operations for Bosch;
- a 2019 → 2021 time hold-out for Bosch;
- leave-one-experiment-out for UMich;
- a time-based split with a 24-hour gap for Azure;
- stratified 5-fold cross-validation for AI4I.

Where a score looks too good, the cause was investigated and reported.

Table 16: Model scorecard (docs/results/ml_*.json)

Model	Data	Validation	ROC-AUC	PR-AUC	Note
Vibration anomaly, Isolation Forest trained on good only	Bosch (REAL)	held-out good + all bad	0.978	0.800	deployable at the edge: needs no fault examples
Vibration anomaly, Random Forest	Bosch (REAL)	stratified 5-fold	0.995	0.986	precision 0.99, recall 0.97
Same, temporal hold-out	Bosch (REAL)	train 2019 → test 2021	0.984	0.930	rules out recording-period leakage (39 of 70 bad files are from Feb 2019)
Same, cross-machine	Bosch (REAL)	train M01 → test M02 / M03	1.00 / 0.91	0.997 / 0.89	M03 is the hardest domain shift, as the dataset paper notes [1]

Model	Data	Validation	ROC-AUC	PR-AUC	Note
Worn vs unworn tool	UMich (REAL)	leave-one-experiment-out	0.36	0.44	**does not generalise** with 18 experiments; a leaky random split gives 0.72
Failed visual inspection	UMich (REAL)	leave-one-experiment-out	0.75	0.30	indicative, n = 14
Component failure within 24 h	Azure PdM (BENCHMARK)	time split Jan–Aug / Sep–Dec, 24-h gap	1.000	0.999	225/225 failures alerted; ablation shows simulated precursors, so a real fleet would score lower
Failure modes HDF / PWF / OSF / TWF	AI4I (BENCHMARK)	stratified 5-fold	1.00 / 1.00 / 1.00 / 0.93	0.99 / 1.00 / 1.00 / 0.09	physics features (ΔT, power, wear × torque) recover the generating rules; TWF is random by design

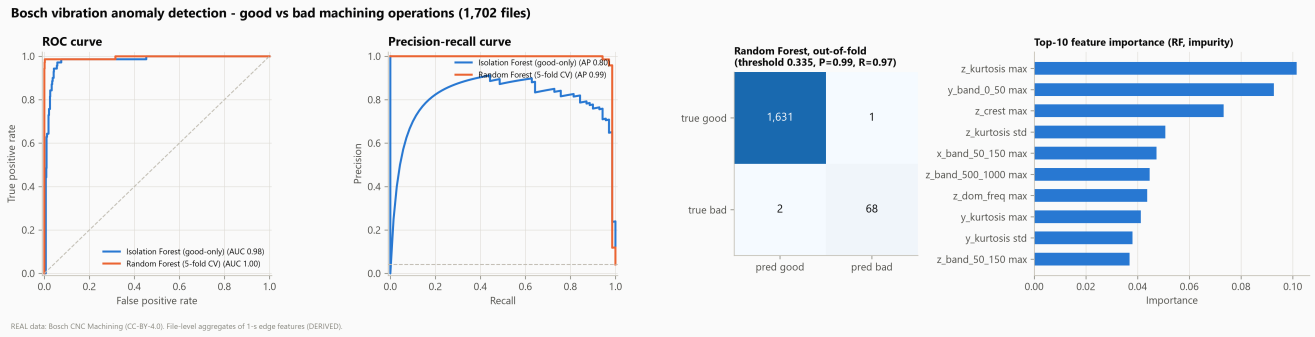


Figure 18: Bosch vibration anomaly detection: ROC, precision-recall, confusion matrix and feature importance

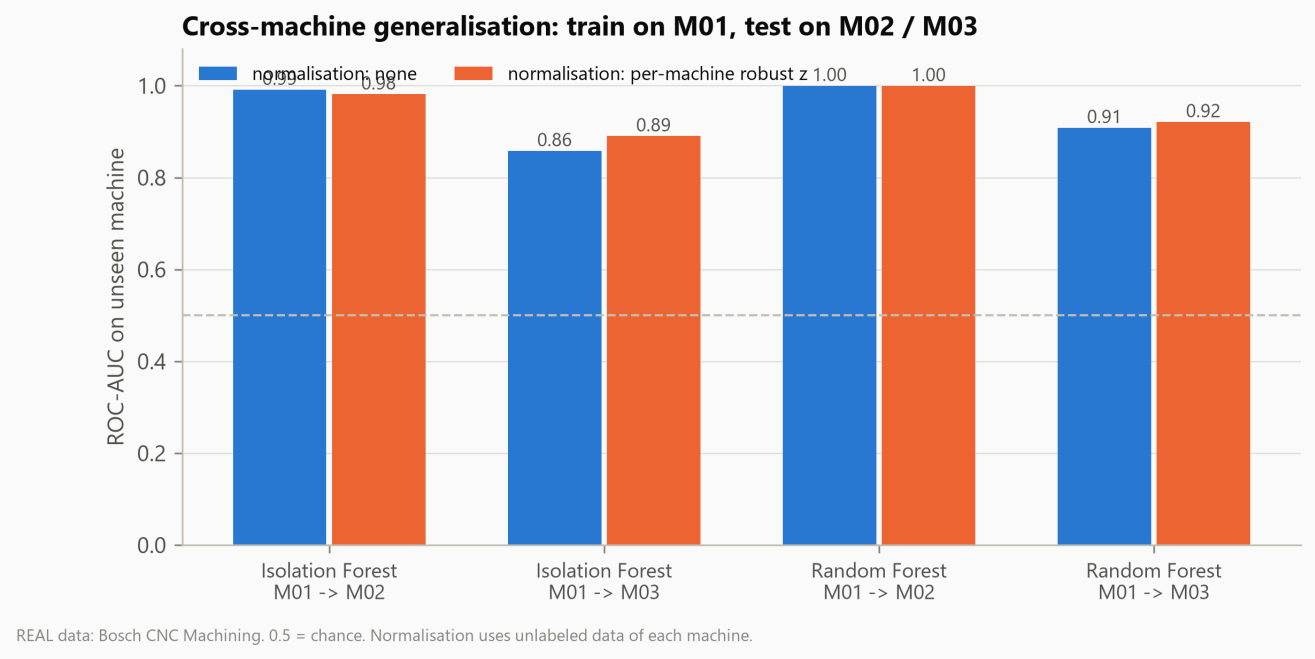


Figure 19: Cross-machine generalisation on Bosch data: train on M01, test on M02 and M03

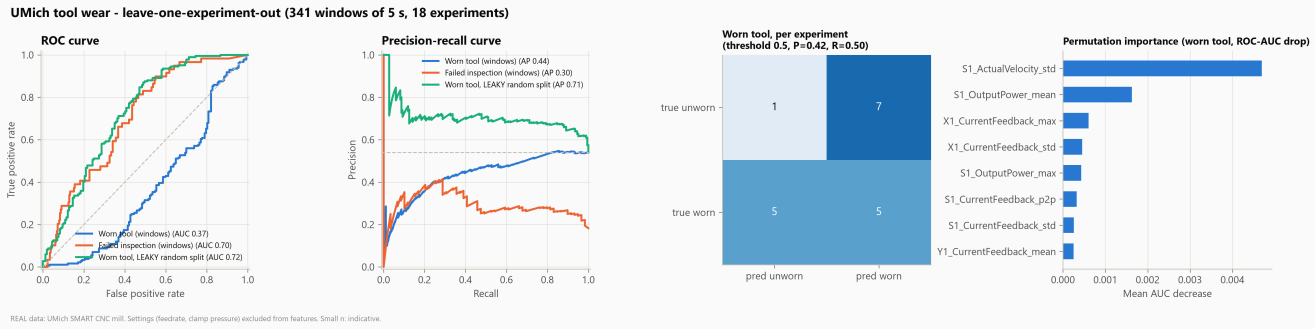


Figure 20: UMich tool wear and inspection outcome, leave-one-experiment-out, including the leaky random split for comparison

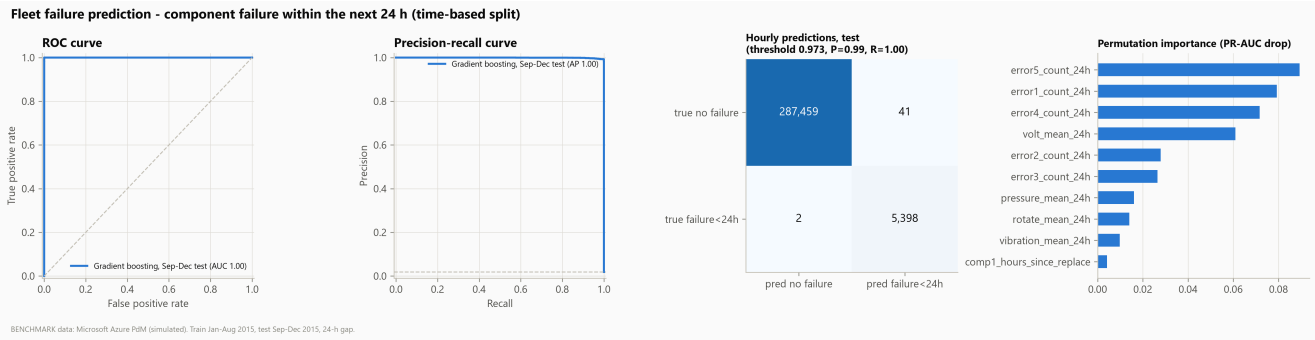


Figure 21: 24-hour failure prediction on the Azure PdM fleet (time-based split)

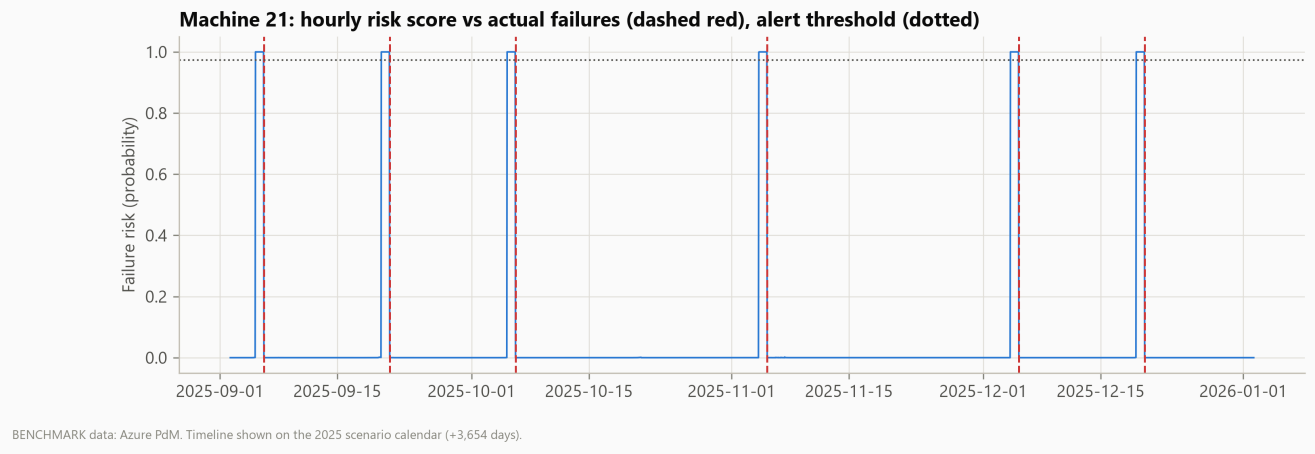


Figure 22: Hourly risk score vs actual failures for the machine with the most test-period failures

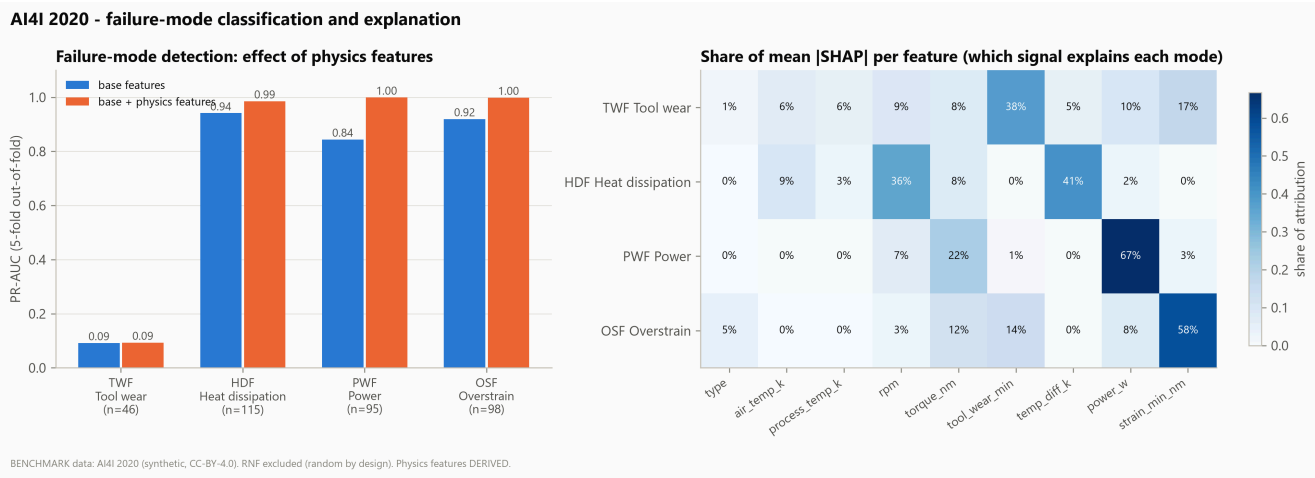


Figure 23: AI4I failure modes: the effect of physics features and the share of mean |SHAP| per feature

Business impact (indicative). The calculation uses the model results and fleet volumes together with stated INR assumptions (`docs/results/business_assumptions.csv`):

- **Downtime cost:** ₹5,000 per machine-hour.
- **Repair premium:** ₹25,000 per unplanned repair.
- **Real-world model performance:** 60% recall and 50% precision.
- **Tool policy:** a wear-based tool change at 150 min, with ₹3,500 per tool.
- **Scrap cost:** 60% of the part value.

Table 17: Indicative annual business impact for the fleet

Lever	Volume effect	Net ₹ per year
Predictive maintenance	446 of 743 failures caught; 2,117 machine-hours of downtime avoided	≈ ₹2.06 crore
Wear-based tool change	Reject rate 1.50% → 0.92%, net of extra tools	≈ ₹3.78 crore
Total		**≈ ₹5.84 crore**

The whole cloud bill (₹4–10 lakh per year) is about 1–2% of this benefit.

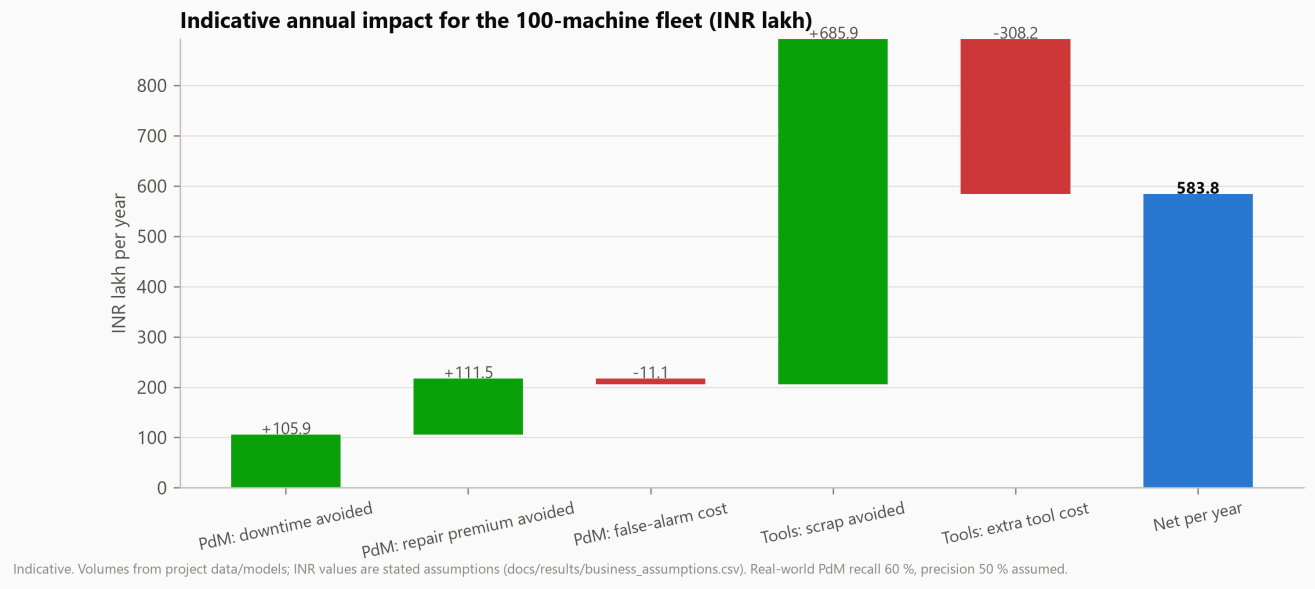


Figure 24: Indicative annual impact for the 100-machine fleet (INR lakh)

10. Dashboard walkthrough

Where it runs.

- **Locally**, the Streamlit dashboard (`dashboard/`) reads PostgreSQL.
- **Hosted** on Streamlit Community Cloud, it reads the bundled Parquet snapshot, so it cannot fail if a database is unavailable on demo day.
- Both modes run the same analysis code. Sidebar filters (plant, date range) apply to every page.

The eight pages are:

1. **Fleet overview:** KPIs (machines running, OEE, parts on the last full day, reject rate, alarms in the last 24 h), OEE components by plant, status mix, daily output.
2. **Plant drill-down:** a shop-floor heatmap (daily running share, or hour-by-hour status), output by part family and shift, and an OEE league table.
3. **Machine health:** vibration with ISO 10816-3 zone bands, temperature, current, tool wear, alarm markers, and a 0–100 health score.

4. **Alarms:** a Pareto with cumulative share, an alarm timeline, alarms per day, and MTBF versus MTTR.
5. **Quality:** reject rate by part family and by machine, and the link to tool wear.
6. **Edge demo:** the real 2 kHz Bosch waveform, 1-s features computed live, a spectrogram, reduction counters, and the MQTT results.
7. **Data volume & cost calculator:** sliders for machines, sensors, Hz, bytes, retention, compression, edge mode and batching. It shows bandwidth, storage and indicative INR per month for each cloud, using the same price file as Part C.
8. **Predictive maintenance:** a risk ranking over time, a per-machine risk timeline against failures, model scorecards, and the INR impact.

[SCREENSHOT: Fleet overview page (local PostgreSQL mode)]

[SCREENSHOT: Machine health page with ISO vibration zones]

[SCREENSHOT: Edge demo page - raw waveform vs features]

[SCREENSHOT: Data volume and cost calculator]

[SCREENSHOT: Predictive maintenance page]

[SCREENSHOT: Hosted dashboard on Streamlit Community Cloud]

11. Limitations and honest discussion

- **The fleet table is derived, not measured.**
  - Its backbone is a *simulated* Microsoft benchmark. Every derived column follows documented, calibrated rules, but absolute values (reject counts, temperatures, current) are illustrative.
  - The real-data results are the Bosch and UMich analyses (edge features, compression, anomaly and tool-wear models).
  - Patterns found in the fleet table (OEE, MTBF, Pareto) demonstrate the method, not a real plant's performance.

- **Temperature.** AI4I's process temperature hardly depends on load (only air temperature matters), so fleet temperature varies mainly with plant ambient and season. No load-heating term was invented.
- **Vibration scaling.** The Bosch accelerometer data is in raw units, so only the *ratio* between bad and good operations was transferred to mm/s. Bosch faults are mostly spectral: the median bad/good RMS ratio is just 1.23.
- **Some scores are too good to generalise.**
  - The Azure 24-h failure model is near-perfect because the simulation injects clear precursors.
  - The UMich worn-tool model fails across experiments; 18 experiments are too few.
  - Both results are reported, not hidden.
- **Costs are indicative.**
  - They use list prices on one date, with no networking, support, reservations, GST or labour.
  - GCP India prices were not retrievable, and the GCP partner edge licence is excluded.
  - The Azure IoT Hub bypass variant is unvalidated.
- **Service-status evidence has gaps.** The Azure IoT Central retirement notice was read through a third-party mirror. A few India-availability facts are marked uncertain in `docs/cloud_research_notes.md`.
- **Scores are judgements.** The scores in the decision matrix are evidence-based but still judgements. The sensitivity analysis bounds how much they matter.

12. Conclusion and future work

The architecture. The factory should adopt an **edge-first architecture on Microsoft Azure (Central India)**:

- IoT Operations reads OPC UA at each plant and turns 2 kHz vibration into 42 features per second, a 143× reduction measured on real data.
- Data is batched one message per machine-second and buffered for 7 days.
- Only about 0.7 Mbps for the whole fleet goes to the cloud. There it feeds Eventhouse and ADLS, PostgreSQL, Azure ML and Power BI.

Why it holds.

- The decision is robust to weighting: Azure wins 88.7% of Monte-Carlo runs.
- It follows from the scenario: brownfield OPC UA, Microsoft tooling already in the plants, a Pune region next to Chakan, and an edge design that removes Azure's weaknesses.

Future work.

1. A **pilot at Chakan** on one HMC line, to validate the OPC UA licences, the IoT Operations footprint and the IoT Hub bypass.
2. **OEE rollout** with real cycle signals from the controllers, replacing derived production counts.
3. **MLOps:** a model registry, a drift monitor for the cross-machine shift, and OTA deployment of the Isolation Forest to the gateways.
4. A **digital twin** of each line in Fabric, for what-if scheduling.
5. Collecting **real labelled failures** in the plants, to retrain the PdM model away from the benchmark.

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Appendix A: Full required-query outputs

Table 18: Full per-machine outputs of Q1 (avg temperature), Q4 (alarms), Q7 (avg RPM running) and Q8 (rejection rate). Source: `docs/results/appendix_per_machine.csv`.

machine_code	plant_code	machine_type	avg_temperature_c	alarm_events	alarm_hours	avg_rpm_running	parts	rejection_rate_pct
P1-HMC-01	P1	HMC	36.98	42	40	5325	28173	1.778
P1-HMC-02	P1	HMC	36.98	32	27	5330	28484	1.208
P1-HMC-03	P1	HMC	36.99	44	40	5340	28367	1.459
P1-HMC-04	P1	HMC	36.99	37	33	5326	28280	1.574
P1-HMC-05	P1	HMC	36.99	45	41	5337	28146	1.734
P1-HMC-06	P1	HMC	36.99	23	23	5367	28432	1.562
P1-HMC-07	P1	HMC	36.99	49	43	5327	28331	1.405
P1-HMC-08	P1	HMC	36.99	47	45	5345	28314	1.427
P1-HMC-09	P1	HMC	36.98	49	44	5317	28276	1.457
P1-HMC-10	P1	HMC	36.98	50	46	5325	28403	1.348
P1-HMC-11	P1	HMC	36.98	45	41	5338	28368	1.347
P1-HMC-12	P1	HMC	36.99	47	42	5347	28329	1.564
P1-HMC-13	P1	HMC	36.99	55	46	5336	52521	1.815
P1-HMC-14	P1	HMC	36.98	42	40	5328	52728	1.553
P1-HMC-15	P1	HMC	36.99	58	52	5327	53069	1.296
P1-HMC-16	P1	HMC	36.97	54	48	5337	52752	1.488
P1-HMC-17	P1	HMC	36.98	64	56	5324	52585	1.765
P1-HMC-18	P1	HMC	36.98	54	51	5333	52906	1.537
P1-HMC-19	P1	HMC	37	35	35	5357	52695	1.681
P1-HMC-20	P1	HMC	36.98	60	56	5336	52680	1.56
P1-HMC-21	P1	HMC	36.98	51	46	5328	53008	1.398
P1-HMC-22	P1	HMC	36.99	75	67	5313	52755	1.293
P1-HMC-23	P1	HMC	36.99	49	45	5330	52895	1.369
P1-HMC-24	P1	HMC	36.99	43	43	5340	52668	1.496
P1-VMC-01	P1	VMC	37	45	38	6894	36054	1.39
P1-VMC-02	P1	VMC	36.99	48	45	6875	36148	1.25
P1-VMC-03	P1	VMC	37	34	32	6881	36089	1.463
P1-VMC-04	P1	VMC	36.99	45	43	6877	36185	1.296
P1-VMC-05	P1	VMC	37.01	45	44	6890	35965	1.526
P1-VMC-06	P1	VMC	36.99	45	42	6885	36054	1.317
P1-VMC-07	P1	VMC	36.99	33	29	6911	36201	1.229
P1-VMC-08	P1	VMC	36.98	47	43	6881	36098	1.405
P1-VMC-09	P1	VMC	36.98	46	41	6863	36004	1.614
P1-VMC-10	P1	VMC	36.99	37	36	6907	36273	1.188
P1-VMC-11	P1	VMC	36.98	51	47	6893	35867	1.695
P1-VMC-12	P1	VMC	36.98	36	34	6885	36181	1.415
P1-VMC-13	P1	VMC	36.98	57	54	6859	35828	1.686
P1-VMC-14	P1	VMC	36.98	49	45	6877	35816	1.739
P1-VMC-15	P1	VMC	37	41	38	6889	36245	1.418
P1-VMC-16	P1	VMC	36.98	51	49	6911	35934	1.386
P2-TURN-01	P2	TURN	39.11	34	34	2308	113704	1.367
P2-TURN-02	P2	TURN	39.11	47	44	2303	113052	1.793
P2-TURN-03	P2	TURN	39.12	48	44	2308	113071	1.655
P2-TURN-04	P2	TURN	39.11	36	34	2300	113625	1.725
P2-TURN-05	P2	TURN	39.11	44	43	2307	113262	1.718
P2-TURN-06	P2	TURN	39.11	31	29	2313	113160	1.883
P2-TURN-07	P2	TURN	39.1	49	44	2291	113350	1.336
P2-TURN-08	P2	TURN	39.12	37	33	2292	113588	1.412

machine_code	plant_code	machine_type	avg_temperature_c	alarm_events	alarm_hours	avg_rpm_running	parts	rejection_rate_pct
P2-TURN-09	P2	TURN	39.1	63	58	2292	113146	1.549
P2-TURN-10	P2	TURN	39.11	47	44	2290	113993	1.401
P2-HOB-01	P2	HOB	39.12	50	45	961	87913	1.69
P2-HOB-02	P2	HOB	39.11	53	47	967	88321	1.5
P2-HOB-03	P2	HOB	39.11	51	48	966	88400	1.502
P2-HOB-04	P2	HOB	39.11	51	49	970	88191	1.543
P2-HOB-05	P2	HOB	39.1	45	43	966	88410	1.348
P2-HOB-06	P2	HOB	39.11	49	44	969	88198	1.569
P2-HOB-07	P2	HOB	39.13	35	35	969	88454	1.51
P2-HOB-08	P2	HOB	39.1	43	41	963	88297	1.515
P2-HOB-09	P2	HOB	39.13	43	41	966	88382	1.672
P2-HOB-10	P2	HOB	39.11	47	43	969	88448	1.36
P2-TURN-11	P2	TURN	39.11	38	35	2301	66490	1.448
P2-TURN-12	P2	TURN	39.1	35	33	2301	66048	1.587
P2-TURN-13	P2	TURN	39.11	58	53	2297	66031	1.683
P2-TURN-14	P2	TURN	39.13	45	42	2290	66304	1.544
P2-TURN-15	P2	TURN	39.11	41	38	2308	66394	1.566
P2-GRIND-01	P2	GRIND	39.12	48	45	3174	158565	1.9
P2-GRIND-02	P2	GRIND	39.11	48	43	3157	157953	1.641
P2-GRIND-03	P2	GRIND	39.12	42	39	3162	159123	1.338
P2-GRIND-04	P2	GRIND	39.11	53	50	3162	157903	1.871
P2-GRIND-05	P2	GRIND	39.12	42	40	3170	158988	1.445
P2-GRIND-06	P2	GRIND	39.11	58	53	3166	157438	1.973
P2-GRIND-07	P2	GRIND	39.11	44	44	3186	159380	1.397
P2-GRIND-08	P2	GRIND	39.11	51	51	3184	157816	1.765
P2-GRIND-09	P2	GRIND	39.11	42	37	3167	159258	1.706
P2-GRIND-10	P2	GRIND	39.11	47	43	3154	157990	1.87
P3-TURN-01	P3	TURN	36.49	41	39	2309	198019	1.248
P3-TURN-02	P3	TURN	36.48	22	22	2308	199491	1.237
P3-TURN-03	P3	TURN	36.48	63	61	2304	196575	1.701
P3-TURN-04	P3	TURN	36.48	60	57	2300	197936	1.426
P3-TURN-05	P3	TURN	36.48	49	45	2298	199677	1.195
P3-TURN-06	P3	TURN	36.49	59	53	2293	198326	1.21
P3-TURN-07	P3	TURN	36.49	37	36	2300	199248	1.158
P3-TURN-08	P3	TURN	36.48	58	50	2289	197575	1.675
P3-TURN-09	P3	TURN	36.49	50	44	2289	197392	1.385
P3-VMC-01	P3	VMC	36.49	51	46	6903	79290	1.251
P3-VMC-02	P3	VMC	36.48	31	29	6877	79502	1.333
P3-VMC-03	P3	VMC	36.46	48	47	6902	79476	1.226
P3-VMC-04	P3	VMC	36.48	60	51	6899	79225	1.382
P3-VMC-05	P3	VMC	36.47	38	35	6867	79205	1.506
P3-VMC-06	P3	VMC	36.48	53	47	6865	78723	1.517
P3-VMC-07	P3	VMC	36.46	48	46	6924	79247	1.615
P3-VMC-08	P3	VMC	36.46	54	49	6905	79446	1.257
P3-TURN-10	P3	TURN	36.46	37	35	2301	131988	1.501
P3-TURN-11	P3	TURN	36.47	55	50	2294	132824	1.231
P3-TURN-12	P3	TURN	36.46	46	44	2303	131535	1.563
P3-TURN-13	P3	TURN	36.49	44	44	2314	132676	1.139
P3-TURN-14	P3	TURN	36.48	53	50	2306	132352	1.466
P3-TURN-15	P3	TURN	36.44	61	53	2297	131676	1.362
P3-TURN-16	P3	TURN	36.45	73	67	2285	130995	1.476
P3-TURN-17	P3	TURN	36.49	34	33	2314	132661	1.326

Table 19: Q8 top-10 output. Source: `sql/results/q8_highest_rejection_rate.csv`.

rnk	machine_code	part_name	produced	rejected	rejection_rate_pct
1	P2-GRIND-06	Gear shaft (finish grind)	157438	3106	1.973
2	P2-GRIND-01	Gear shaft (finish grind)	158565	3013	1.9
3	P2-TURN-06	Gear shaft	113160	2131	1.883
4	P2-GRIND-04	Gear shaft (finish grind)	157903	2954	1.871
5	P2-GRIND-10	Gear shaft (finish grind)	157990	2954	1.87
6	P1-HMC-13	Cylinder head	52521	953	1.815
7	P2-TURN-02	Gear shaft	113052	2027	1.793
8	P1-HMC-01	Engine block	28173	501	1.778
9	P1-HMC-17	Cylinder head	52585	928	1.765
10	P2-GRIND-08	Gear shaft (finish grind)	157816	2785	1.765

Table 20: A4 Shift-wise production. Source: `sql/results/a4_shift_production.csv`.

plant_code	shift	parts	rejects	reject_rate_pct	running_hours	alarm_hours	share_of_plant_output_pct
P1	A	505751	7485	1.48	88489	929	32.63
P1	B	521390	8165	1.566	91197	377	33.64
P1	C	522966	7334	1.402	91529	404	33.74
P2	A	1284707	20932	1.629	77552	804	32.67
P2	B	1322061	22242	1.682	79818	341	33.62
P2	C	1325878	20173	1.521	80041	344	33.71
P3	A	1133807	15322	1.351	55240	626	32.63
P3	B	1168749	17247	1.476	56941	246	33.63
P3	C	1172504	15066	1.285	57195	261	33.74

Appendix B: Assumptions

Table 21: Business-impact assumptions (INR). Source: `docs/results/business_assumptions.csv`.

assumption	value
downtime_cost_inr_per_machine_hour	5,000
unplanned_repair_premium_inr	25,000
planned_intervention_hours	3
false_alarm_cost_inr	2,500
realistic_event_recall	0.6
realistic_precision	0.5
tool_change_policy_wear_min	150
tool_cost_inr	3,500
scrap_cost_share_of_part_value	0.6
part_value_inr: Engine block	9,000
part_value_inr: Cylinder head	6,500
part_value_inr: Crankcase	5,000
part_value_inr: Gear shaft	1,200
part_value_inr: Helical gear	900
part_value_inr: Differential housing	2,500
part_value_inr: Gear shaft (finish grind)	1,500

assumption	value
part_value_inr: Brake disc	700
part_value_inr: Steering knuckle	1,800
part_value_inr: Wheel hub	1,100

Table 22: Machine-type assumptions. Source: `data/reference/machine_type.csv`.

machine_type	description	rpm_min	rpm_max	rated_power_kw	supply_voltage_v	power_factor	rated_current_a
HMC	Horizontal machining centre	2000	8000	22	415	0.85	36
VMC	Vertical machining centre	3000	10000	15	415	0.85	24.6
TURN	CNC turning centre	800	3500	18.5	415	0.85	30.3
HOB	CNC gear hobbing machine	300	1500	11	415	0.85	18
GRIND	CNC grinding machine	1500	4500	15	415	0.85	24.6

Table 23: Part-family assumptions. Source: `data/reference/part_master.csv`.

part_family_id	part_code	part_name	plant_id	machine_type	cycle_time_min	cycle_time_low_min	cycle_time_high_min	material
1	PF01	Engine block	1	HMC	14	12	18	Cast iron
2	PF02	Cylinder head	1	HMC	7.5	6	9	Aluminium alloy
3	PF03	Crankcase	1	VMC	11	9	14	Aluminium alloy
4	PF04	Gear shaft	2	TURN	3.5	3	4	Case-hardening steel
5	PF05	Helical gear	2	HOB	4.5	3.5	5.5	Case-hardening steel
6	PF06	Differential housing	2	TURN	6	5	7	Ductile iron
7	PF07	Gear shaft (finish grind)	2	GRIND	2.5	2	3	Case-hardening steel
8	PF08	Brake disc	3	TURN	2	1.5	2.5	Grey cast iron
9	PF09	Steering knuckle	3	VMC	5	4	6	Ductile iron
10	PF10	Wheel hub	3	TURN	3	2.5	3.5	Forged steel

Table 24: Alarm master (illustrative mapping of benchmark events). Source: `data/reference/alarm_master.csv`.

alarm_code	alarm_name	severity	source_event	dominant_signal	dominant_z	n_events
AL-1010	Spindle drive overcurrent / overload	WARNING	error1	volt	0.232	1010
AL-2010	Spindle speed deviation (actual < command)	WARNING	error2	rotate	-0.372	988
AL-2020	Axis following error (servo lag)	WARNING	error3	rotate	-0.454	838
AL-3010	Hydraulic pressure abnormal (tool clamp)	WARNING	error4	pressure	0.382	727
AL-4010	High spindle vibration	WARNING	error5	vibration	0.886	356
AL-9100	Spindle motor / drive failure	CRITICAL	comp1	volt	1.287	192
AL-9200	Spindle gearbox / belt failure	CRITICAL	comp2	rotate	-1.467	259
AL-9300	Hydraulic unit / tool-clamp failure	CRITICAL	comp3	pressure	2.115	131

alarm_code	alarm_name	severity	source_event	dominant_signal	dominant_z	n_events
AL-9400	Tool breakage / spindle bearing damage	CRITICAL	comp4	vibration	1.805	179